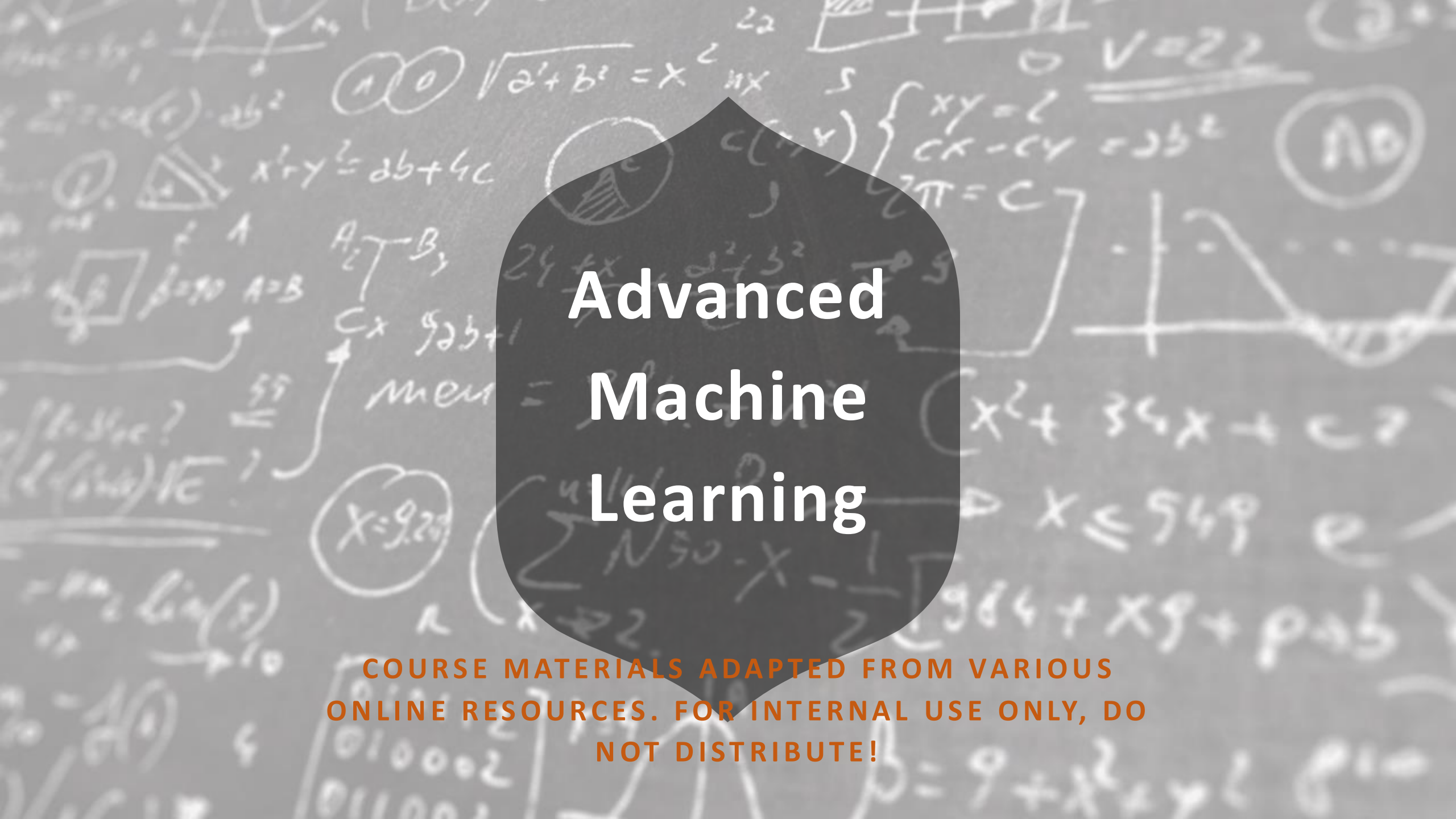


Advanced Machine Learning

COURSE MATERIALS ADAPTED FROM VARIOUS ONLINE RESOURCES. FOR INTERNAL USE ONLY, DO NOT DISTRIBUTE!

Course Logistics

Class Information

Instructor: Songlin Xu, COMP
Office: PQ736
Phone: 2766-7307
Email: songlin.xu@polyu.edu.hk

Teaching Assistant (TA): Pengxiang Li
Email: pengxiang.li@connect.polyu.hk

Office Hour: Wednesday 1pm-3pm
Zoom Link:
https://polyu.zoom.us/j/84823737689
Password: 000000

Class Time

Lecture Place: R503

9:30 am – 12:20 pm, 2:00 pm – 5:20 pm

Week 2 (12 Sep);
Week 5 (3 Oct);
Week 7 (17 Oct);
Week 8 (24 Oct);
Week 10 (7 Nov);
Week 13 (28 Nov).

About me

Songlin Xu

- Assistant Professor,
- Department of Computing,
- The Hong Kong Polytechnic University.

- Ph.D. from University of California San Diego;
- B.Eng. from University of Science and Technology of China.

My research interest

Human-AI Integration System, Internet of Things (IoT), Scientific Discovery.

- **Human Augmentation from AI and Internet of Things (IoT)**
  - Intelligent educational systems to augment student learning
  - Wearable systems for human-AI co-experimentation in wet labs
- **IoT Foundation Models for Health**
  - Sensor foundation models for disease diagnosis
- **Big IoT Data for Science**

Goals of this course

- **Learn about advanced/cutting-edge topics in machine learning.**

- **Practice giving scientific presentations.**

- **Dive deeper into a topic of your choice.**
  - Identifying emerging trends and promising directions.
  - Applying promising techniques to solve your research/real-world problems.

Grading

Coursework	
Research Paper Presentation (in group)	30%
1 Individual Final Project	70%
Total	**100%**

You are expected to present in and regularly attend the lectures based on PolyU Policy.

Advanced topics

- **Topics**
  - Area 1: Pre-training, self-supervised learning, contrastive learning
  - Area 2: Transfer learning, zero/few-shot learning, meta learning
  - Area 3: Deep generative models, VAE, GAN
  - Area 4: Graph-based learning, graph neural networks
  - Area 5: Reinforcement learning

Course organization

- **Weeks 2:**
  - Overview of machine learning paradigms.
- **Weeks 5:**
  - Morning: Area 1 + Group 1 Presentation in Area 1
  - Afternoon: Area 2 + Group 2 Presentation in Area 2
- **Week 7:**
  - Area 3 + Group 3 Presentation in Area 3 (morning: 1 person, afternoon: 1 person)
- **Weeks 8:**
  - Area 4 + Group 4 Presentation in Area 4 (morning: 1 person, afternoon: 1 person)
- **Weeks 10:**
  - Area 5 + Group 5 Presentation in Area 5 (morning: 1 person, afternoon: 1 person)
- **Week 13:**
  - Presentation and submission of individual final project.

Group Area Selection (Scan QR Code for Link)

- You will present one existing research article related to the area you select.
- You are free to select any published research articles based on your interest.



Your name *

Short answer text

Your email *

Short answer text

Your preferred area for presentation (please select at least two or three areas) *

- [ ] Area 1: Pre-training, self-supervised learning, contrastive learning
- [ ] Area 2: Transfer learning, zero/few-shot learning, meta learning
- [ ] Area 3: Deep generative models, VAE, GAN
- [ ] Area 4: Graph-based learning, graph neural networks
- [ ] Area 5: Reinforcement learning

Final individual project

- You are encouraged to choose a topic relevant to your research/development.
- Write a report paper.
- Implementation of a machine learning or a comprehensive survey.
- Presentation and submission due in Nov 25th.

Format and Requirements

- **Format:** submit your report and slides in PDF format on Canvas.
  - Your report should be in the style of a machine learning article (preferably in NeurIPS format). The length of your report *should not exceed* **15 pages (single column with font size 11)** excluding references.

- **Grading: 20** (presentation) + **50** (report) **= 70 points** in total.
  - Presentation: clarity of problem statement; demonstration of your understanding of the problem, your work & contribution; Q&A performance; time management.
  - Report: quality of literature review; in-depth discussion of your thoughts, solutions, and analysis; comprehensiveness of experiments/survey.

Overview of Machine Learning Paradigms

What is Machine Learning?

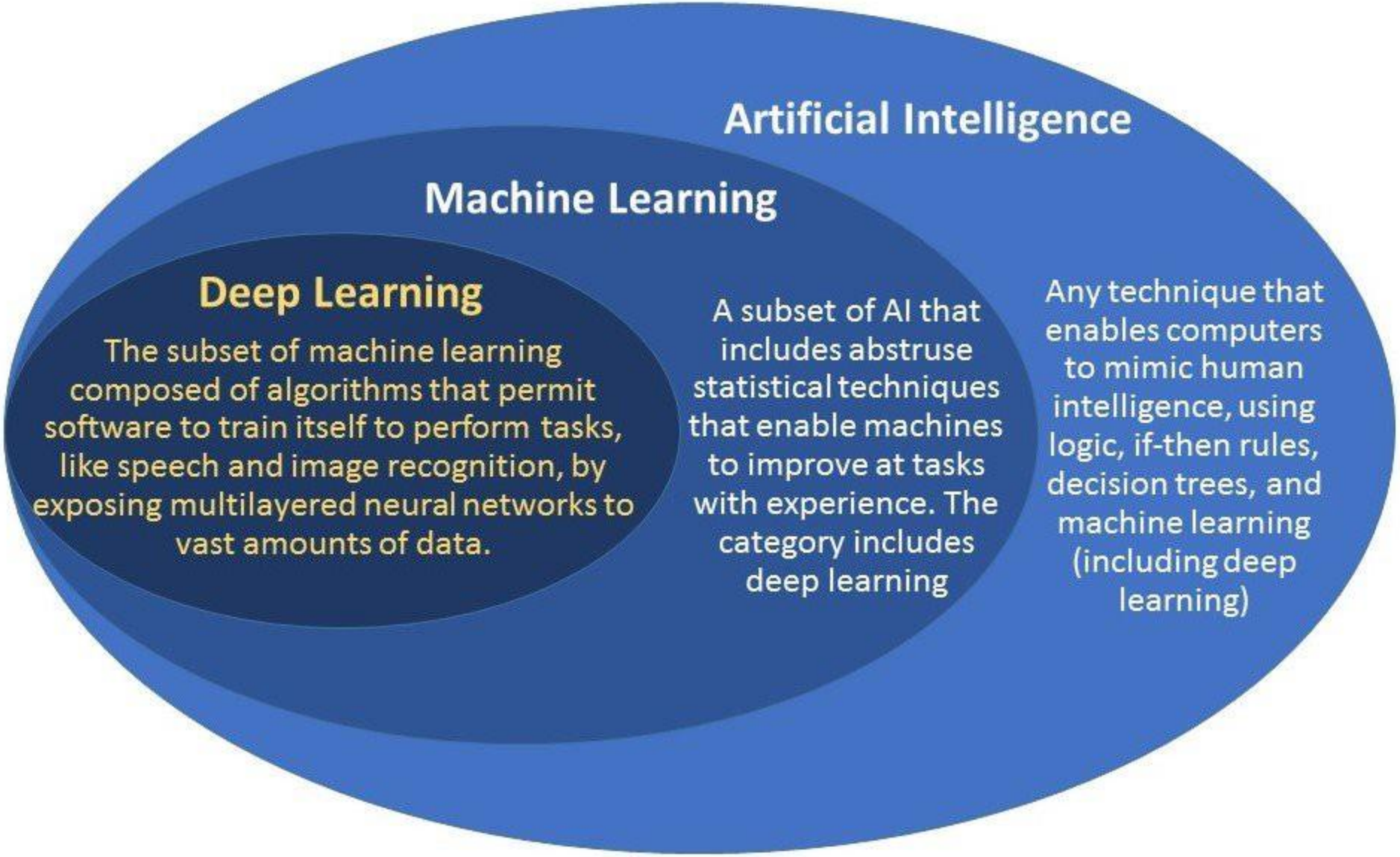


www.educba.com

What is machine learning?

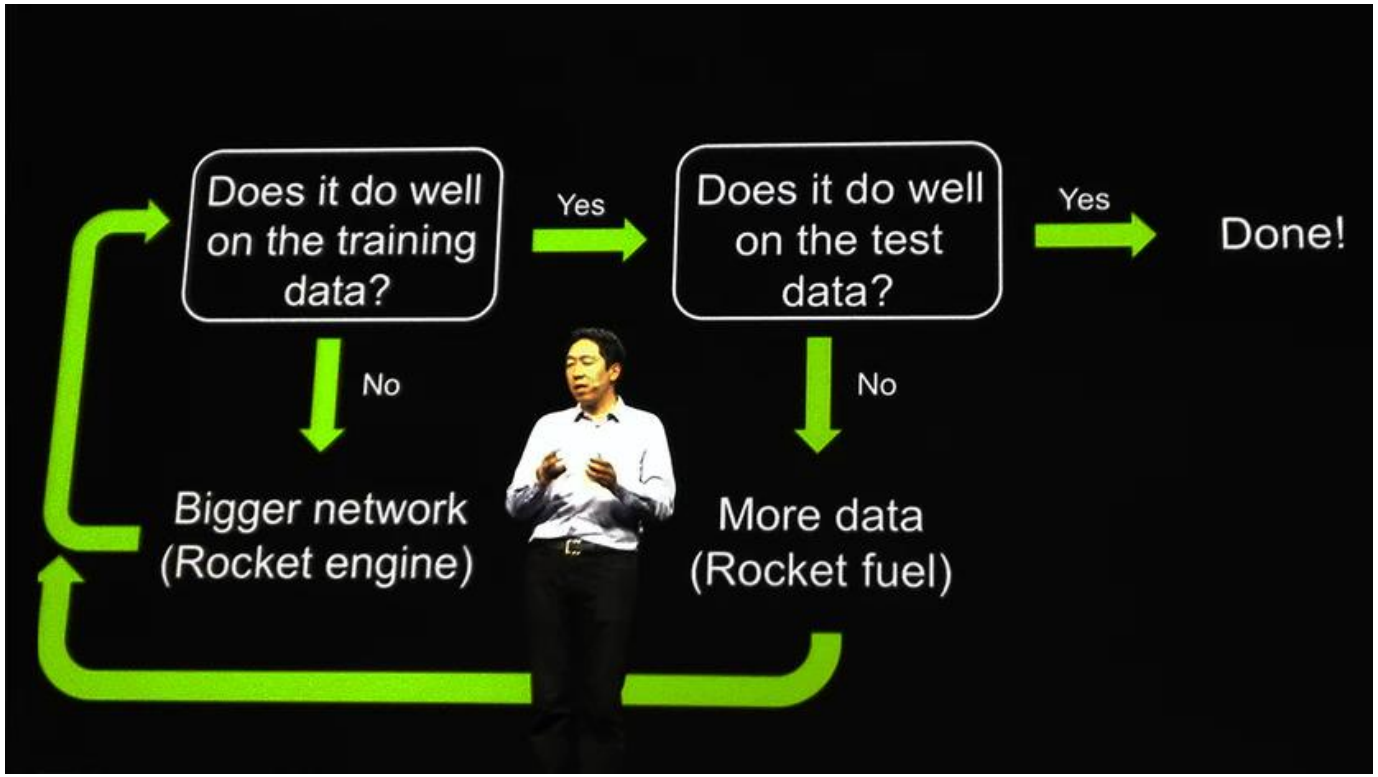
- **Textbook definitions of “machine learning”:**
  - Detecting patterns and regularities with a good and generalizable approximation (“model” or “hypothesis”)
  - Execution of a computer program to optimize the parameters of the model using training data or past experience.

“Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed” - Arthur L. Samuel, 1959



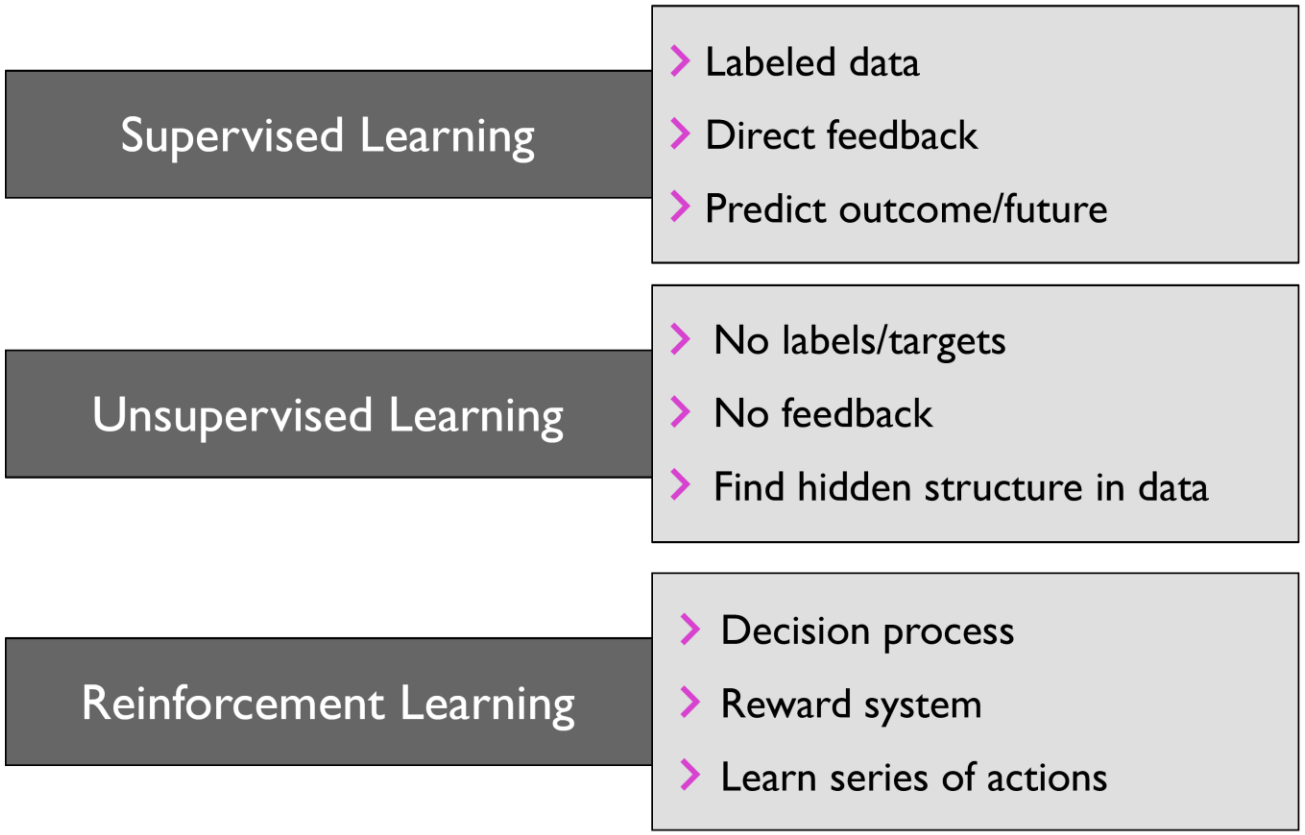
What Is Machine Learning & Deep Learning? | by Claire D. Costa | Medium

The basic recipe for machine learning explained in a single PowerPoint slide



A highly simplified explanation

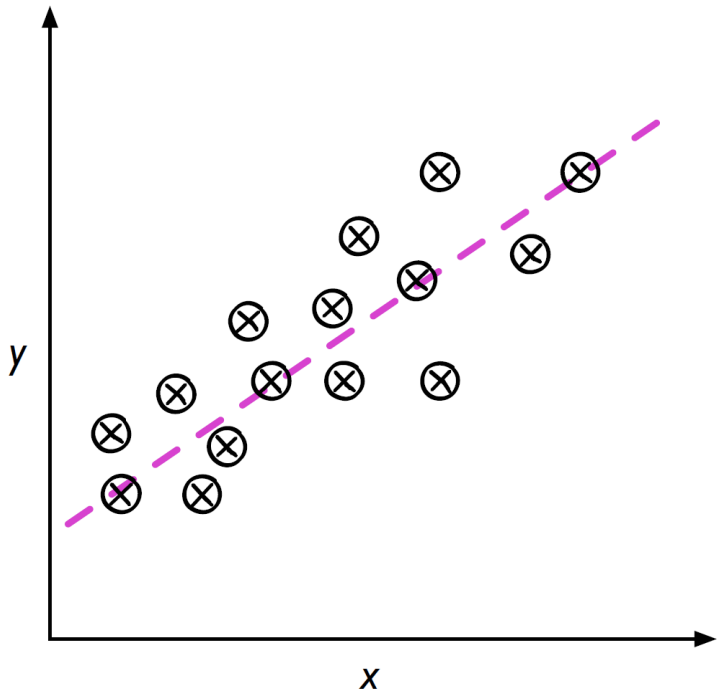
The 3 Broad Categories Of ML (And DL)



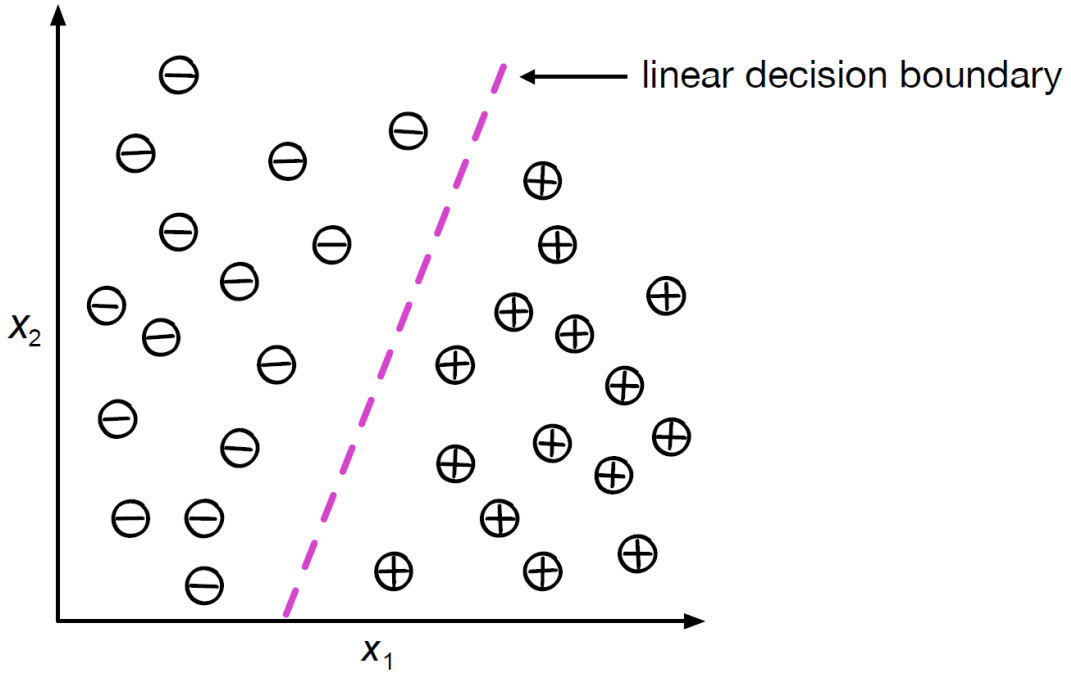
Source: Raschka and Mirjalily (2019). *Python Machine Learning, 3rd Edition*

Supervised Learning

Supervised learning is the largest subcategory



feature (input, observation)

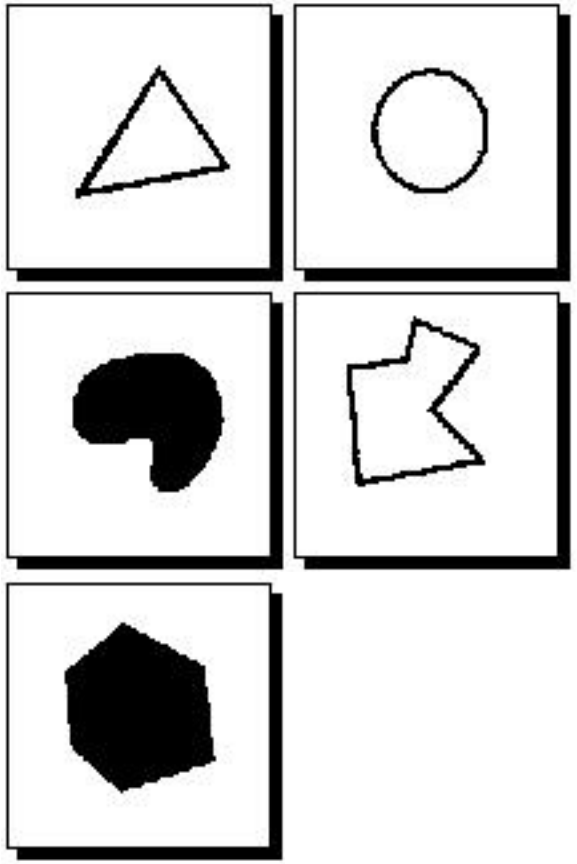


Supervised learning is the largest subcategory

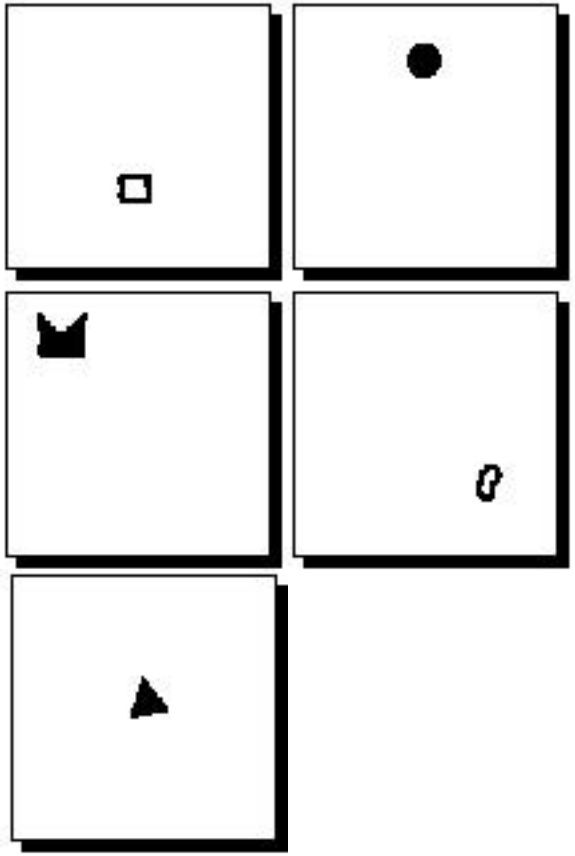
- **Classification**
  - Output is one of a number of classes (e.g., ‘A’)

- **Regression**
  - Output is a real value (e.g., ‘$35/share”)

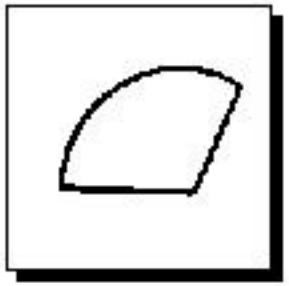
Training Examples: Class 1



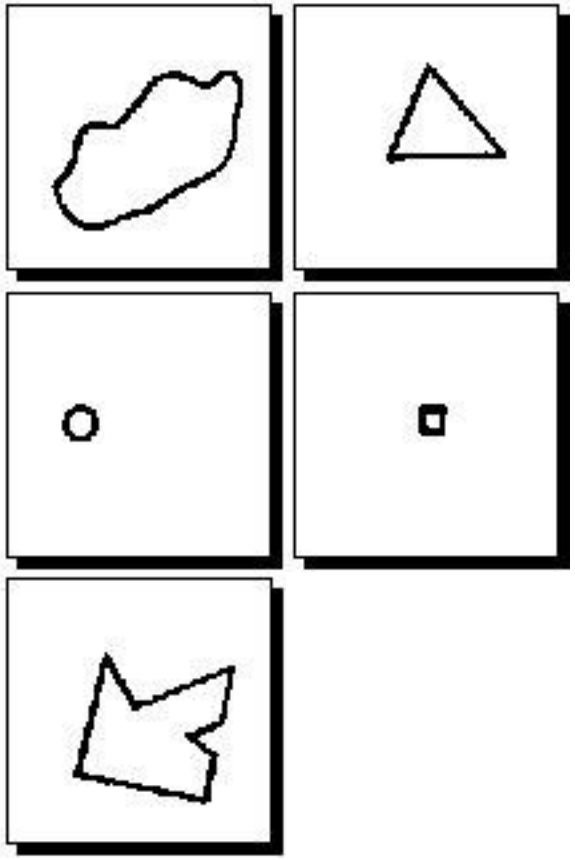
Training Examples: Class 2



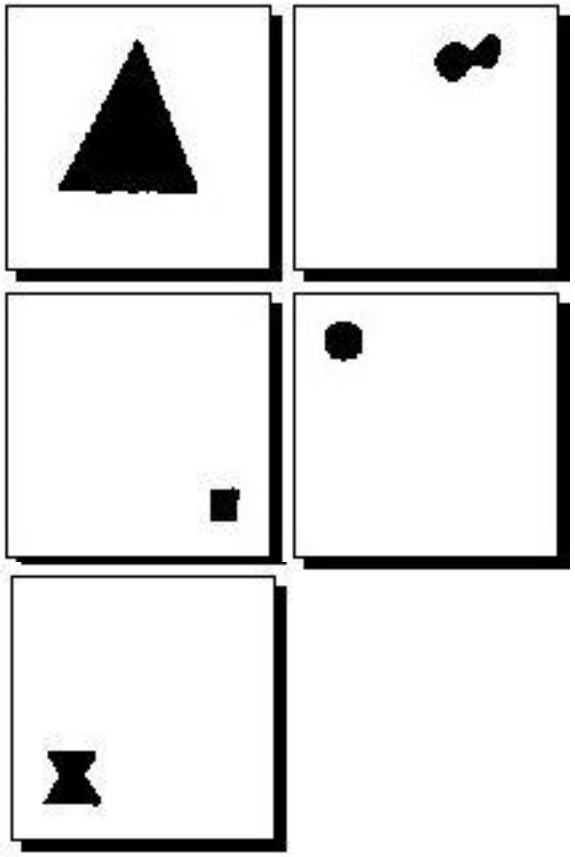
Test example: Class = ?



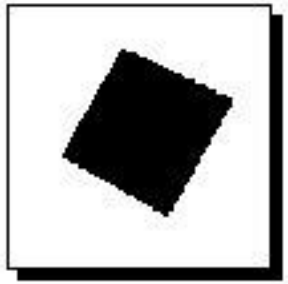
Training Examples: Class 1



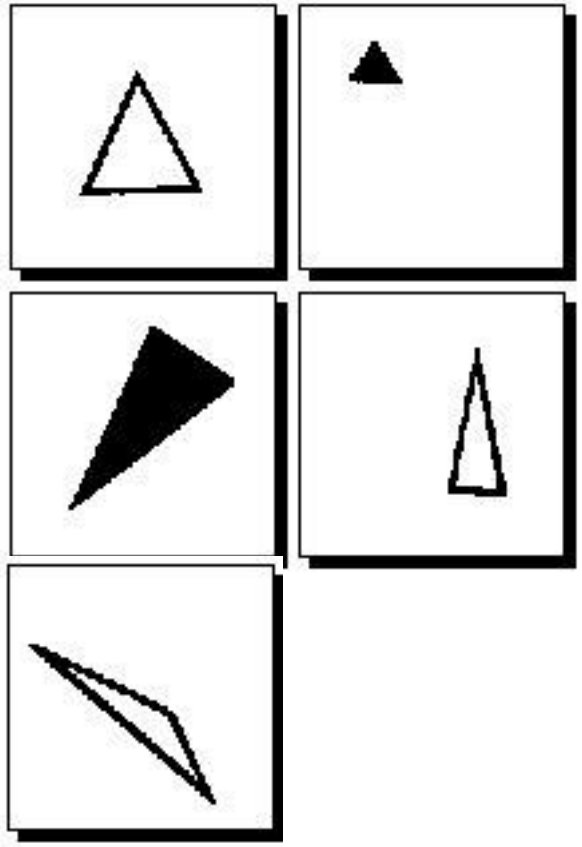
Training Examples: Class 2



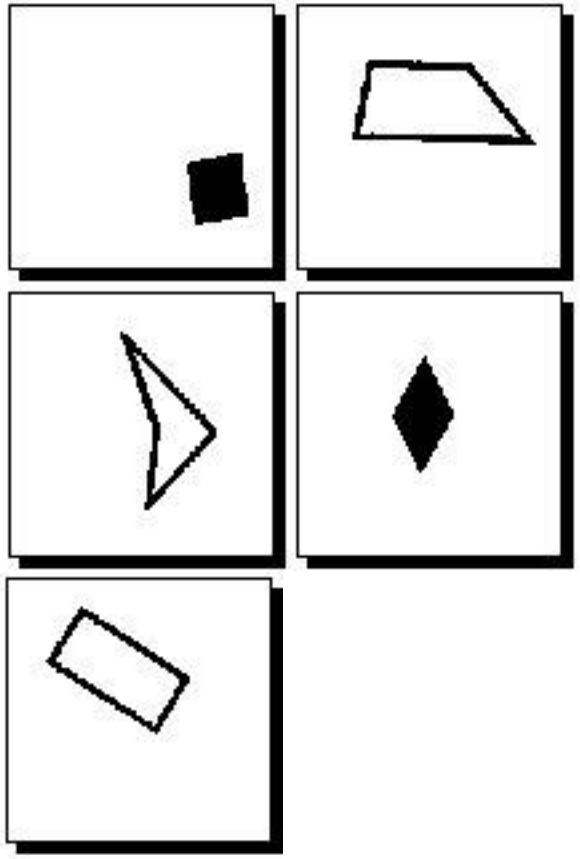
Test example: Class = ?



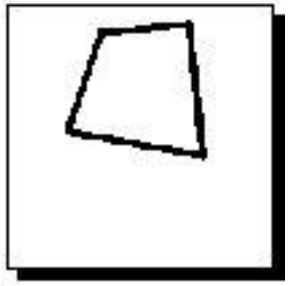
Training Examples: Class 1



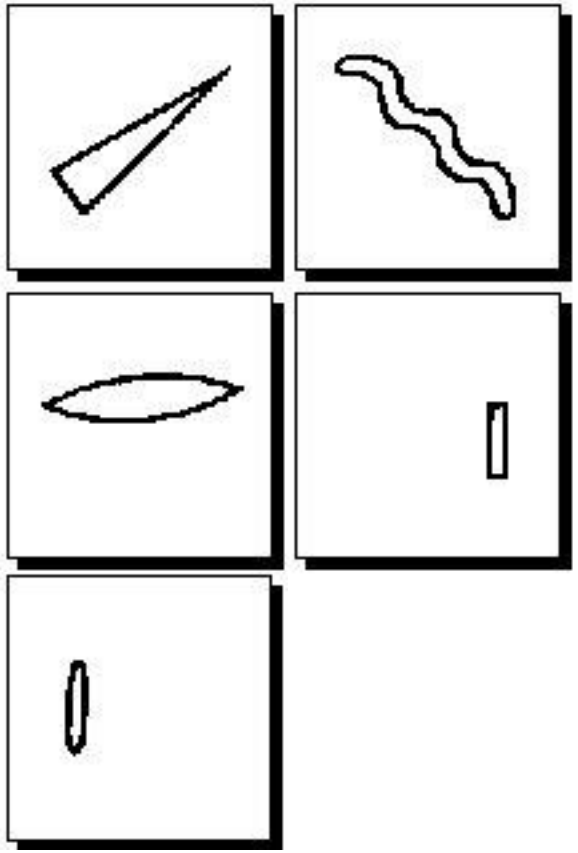
Training Examples: Class 2



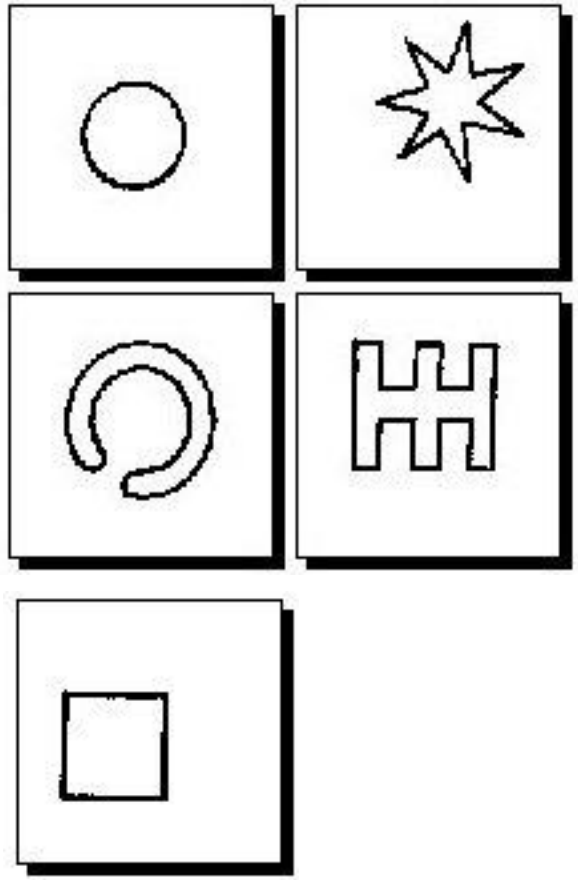
Test example: Class = ?



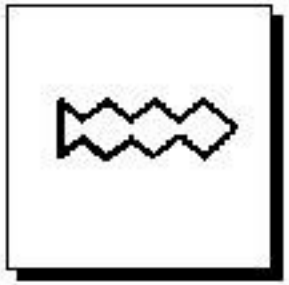
Training Examples: Class 1



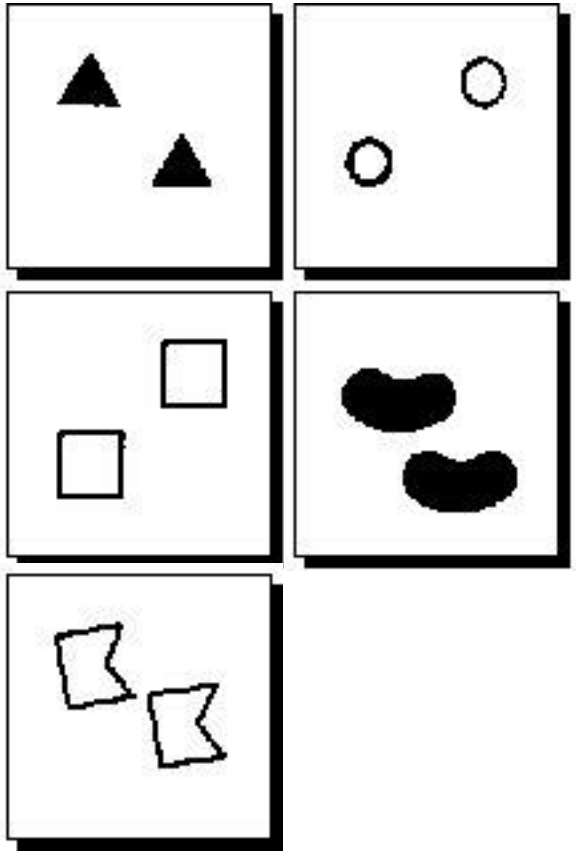
Training Examples: Class 2



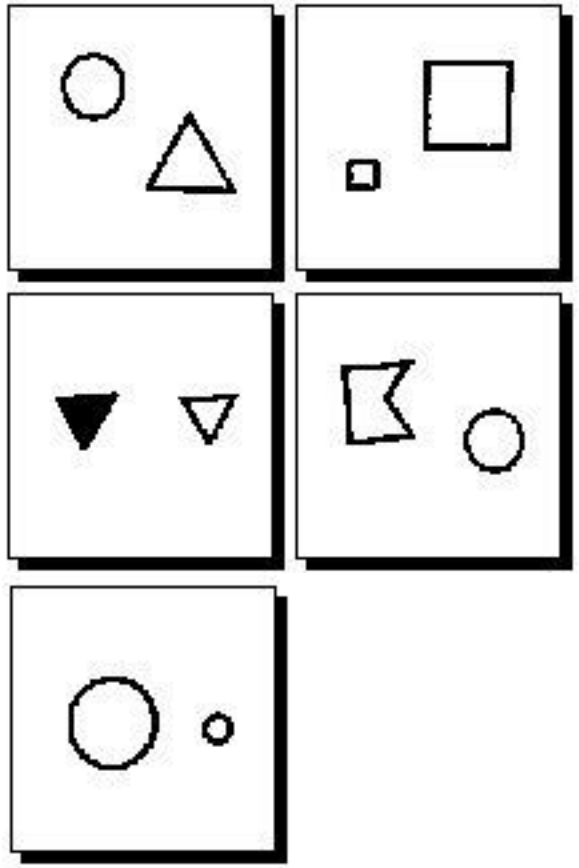
Test example: Class = ?



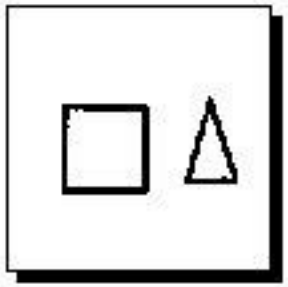
Training Examples: Class 1



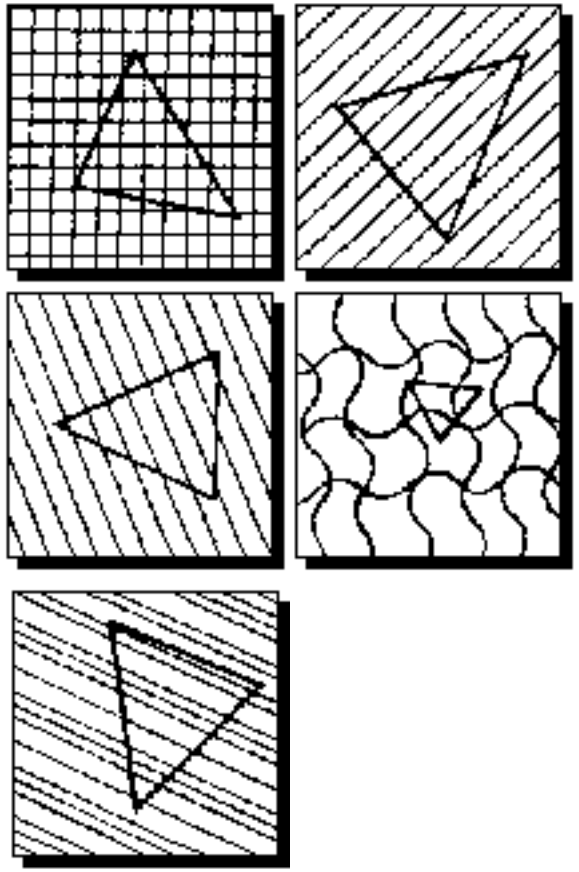
Training Examples: Class 2



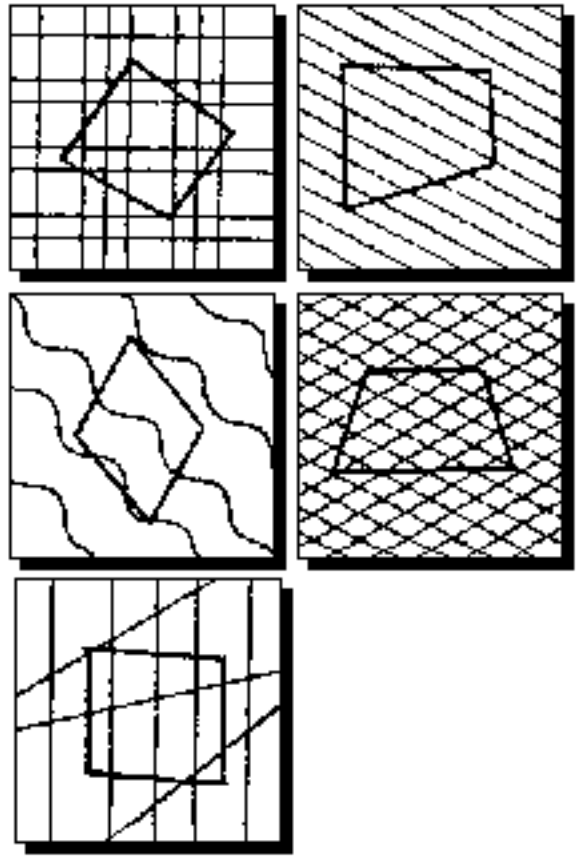
Test example: Class = ?



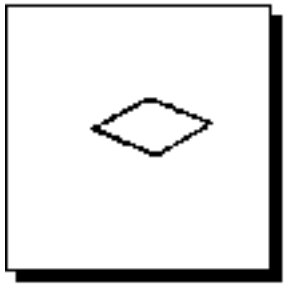
Training Examples: Class 1



Training Examples: Class 2



Test example: Class = ?



Key ingredients for any machine learning method

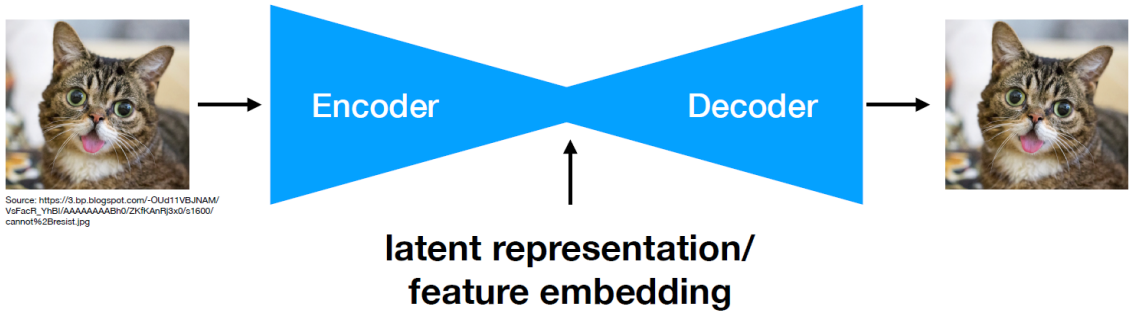
- Features (or “attributes”)
- Underlying Representation for “hypothesis”, “model”, or “target function”
- Hypothesis space
- Learning method
- Data:

Training data	Validation (or Development) data	Test data
Used to train the model	Used to select model hyperparameters, to determine when to stop training, or to alter training method	Used to evaluate trained model

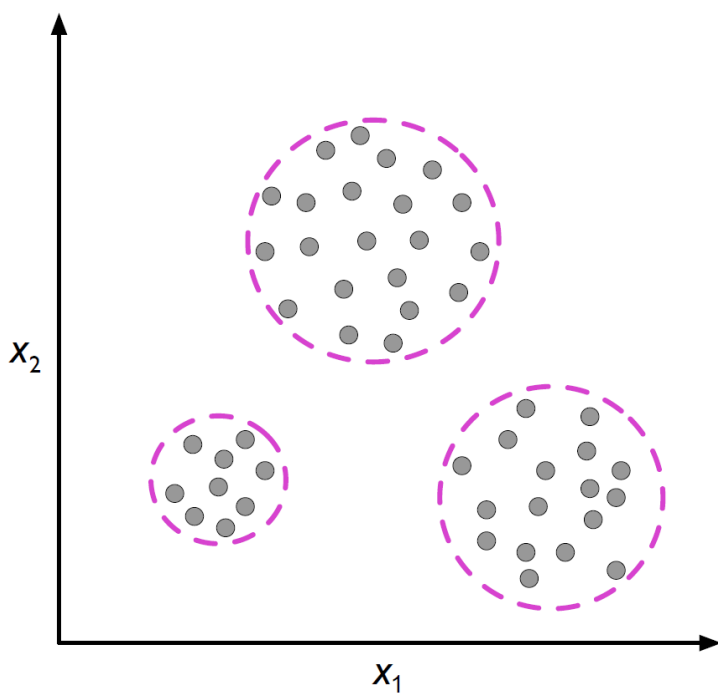
- Evaluation method

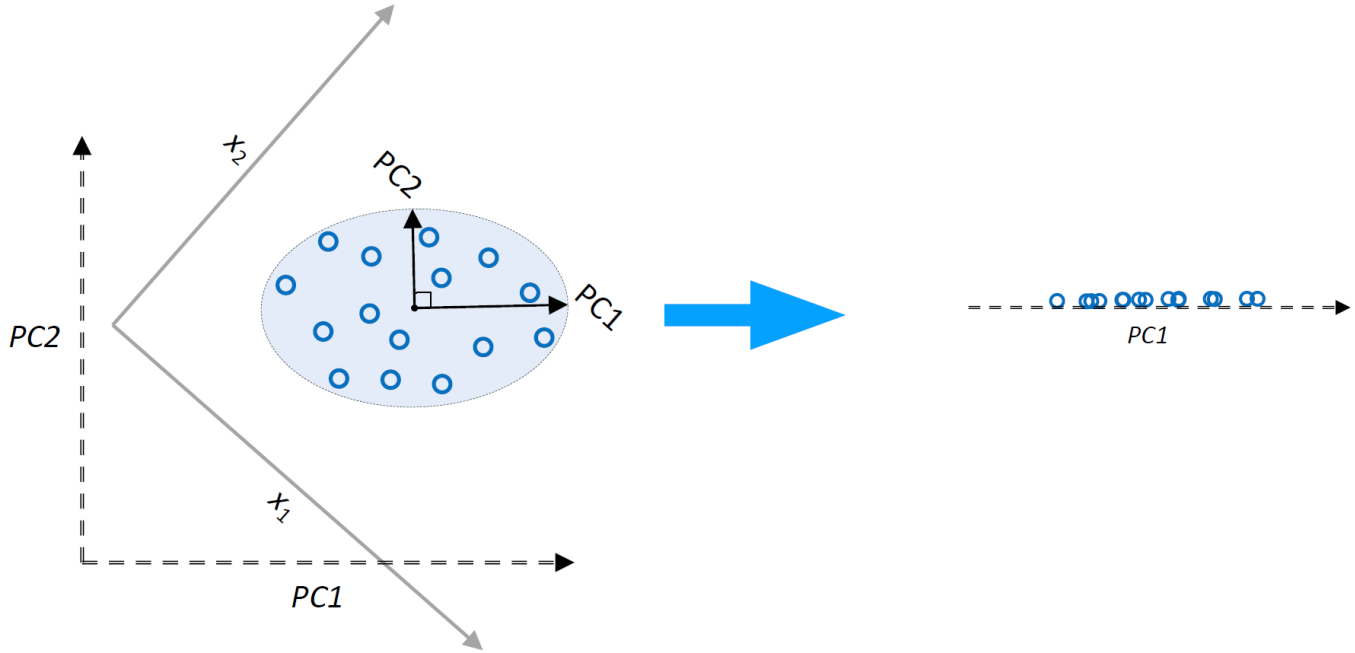
Unsupervised Learning

Unsupervised Learning



Source: https://3.bp.blogspot.com/-OUd11VBJNAM/VsFacR_YhBI/AAAAAAAABh0/ZKfKAnRj3x0/s1600/cannot%2Bresist.jpg





Reinforcement Learning

Examples of reinforcement learning



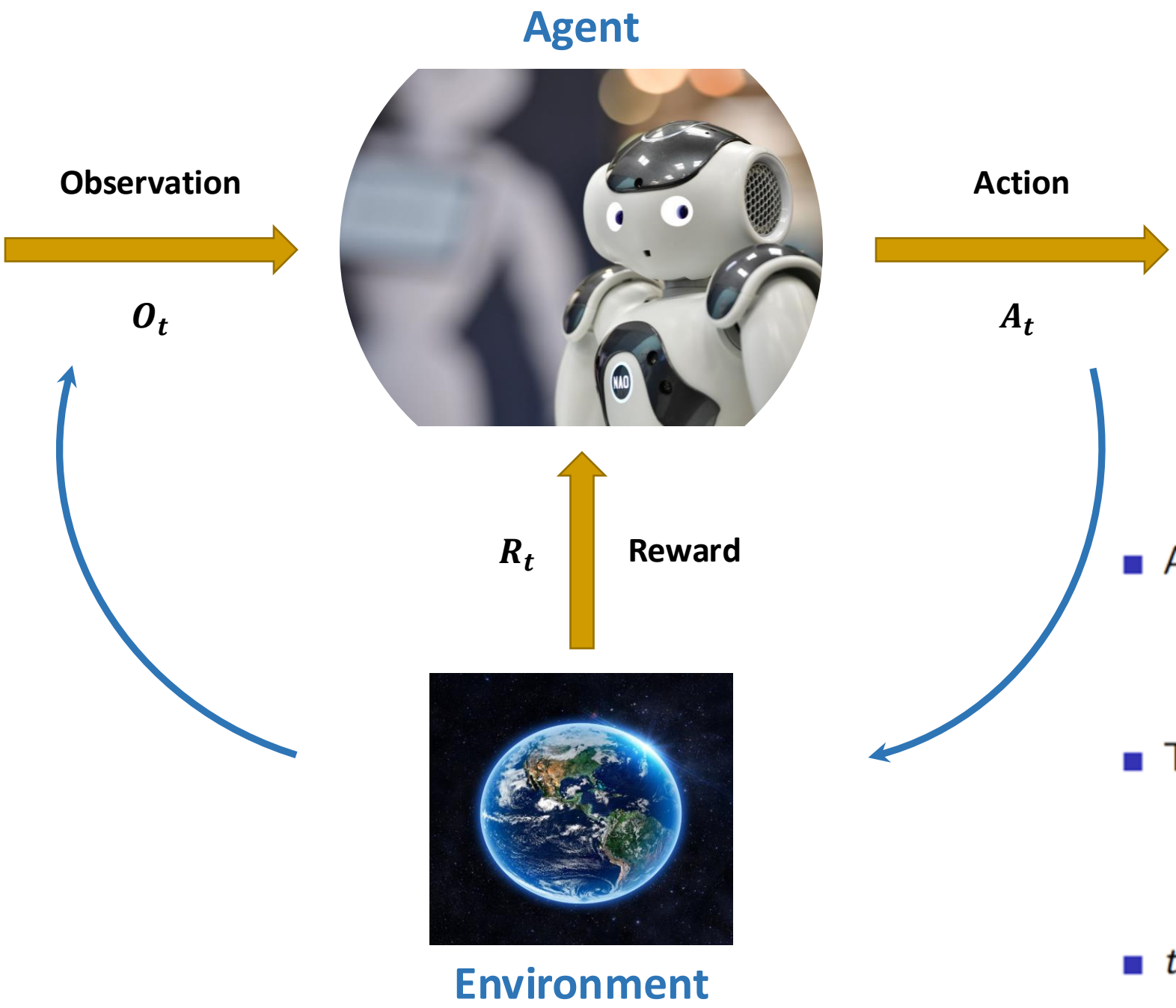
(https://www.youtube.com/watch?v=0JL04JJjocc)

- Fly stunt maneuvers in a helicopter
- Defeat the world champion at Backgammon
- Manage an investment portfolio
- Control a power station
- Make a humanoid robot walk
- Play many different Atari games better than humans

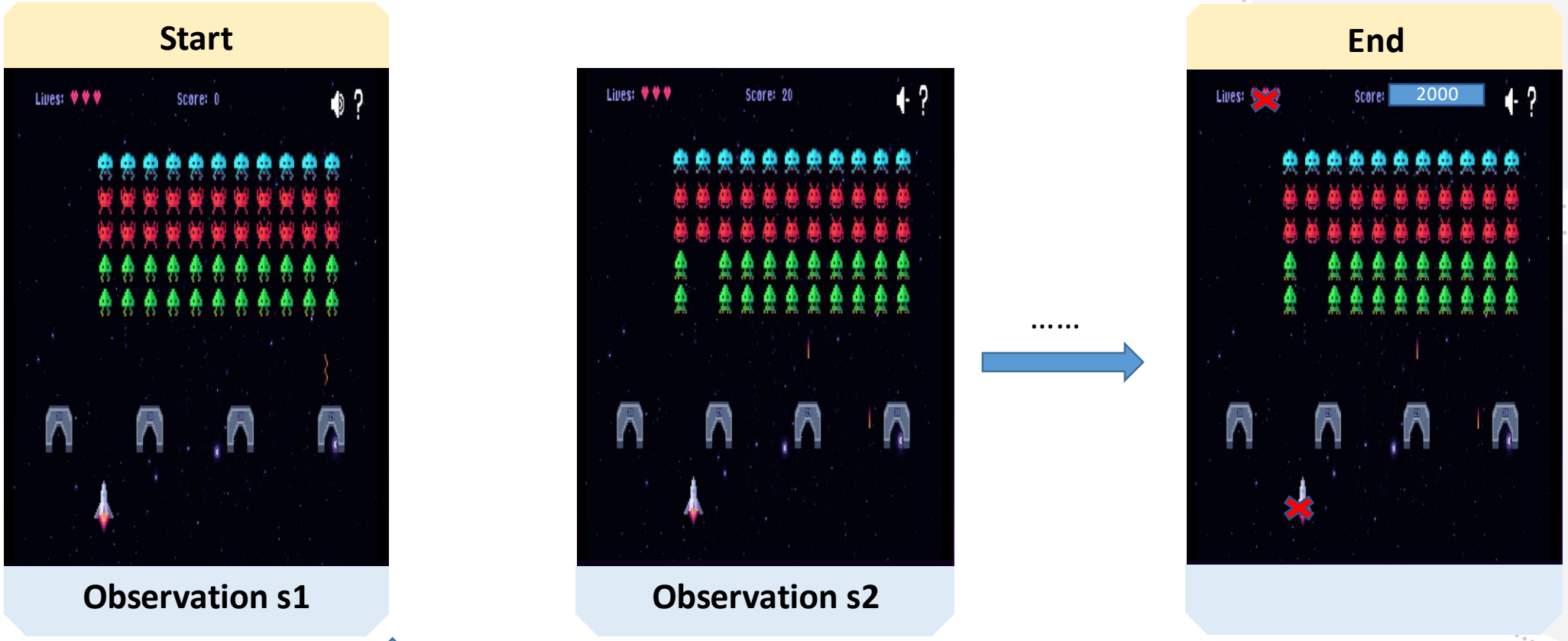
Reinforcement learning

What makes reinforcement learning different from other machine learning paradigms?

- There is no supervisor, only a reward signal
- Feedback is delayed (usually relating to a time sequence), not instantaneous
- Time really matters (sequential)
- Agent’s actions affect the subsequent data it receives



- At each step t the agent:
  - Executes action A_t
  - Receives observation O_t
  - Receives scalar reward R_t
- The environment:
  - Receives action A_t
  - Emits observation O_{t+1}
  - Emits scalar reward R_{t+1}
- t increments at env. step



From start to end: an episode.

Objective: Learn to maximize the expected cumulative reward per episode.

Problems

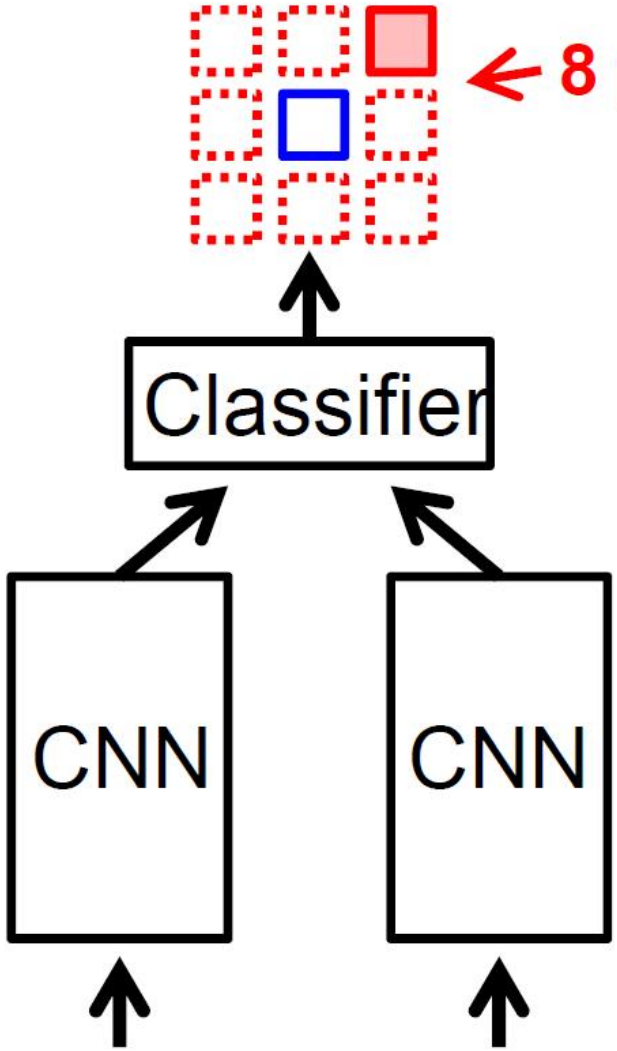
- Actions may have long term consequences
- Reward may be delayed
- It may be better to sacrifice immediate reward to gain more long-term reward
- The interactions between agents and environments are expensive in some cases.
- ……

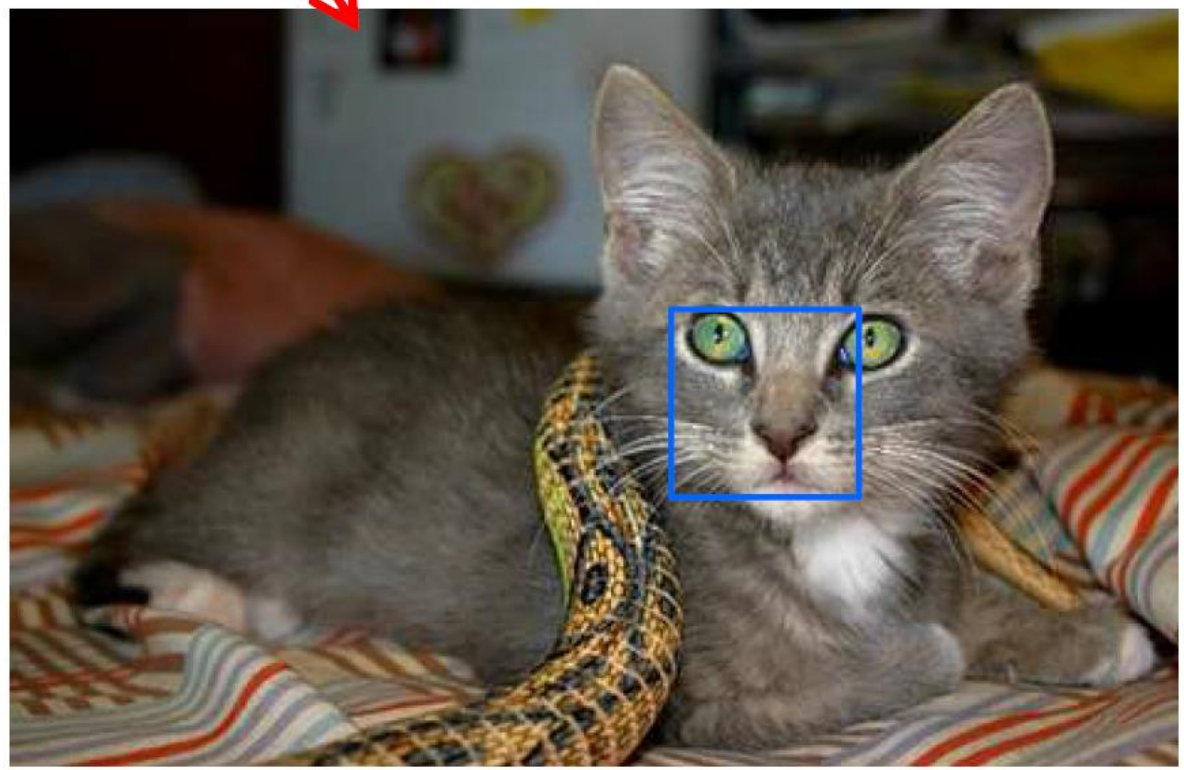
Self-supervised Learning

Self-supervised learning

- Are there alternatives to strong supervision for training?
- **Why self-supervised Learning?**
- **What is Self-Supervision?**

Train network to predict relative position of two regions in the same image



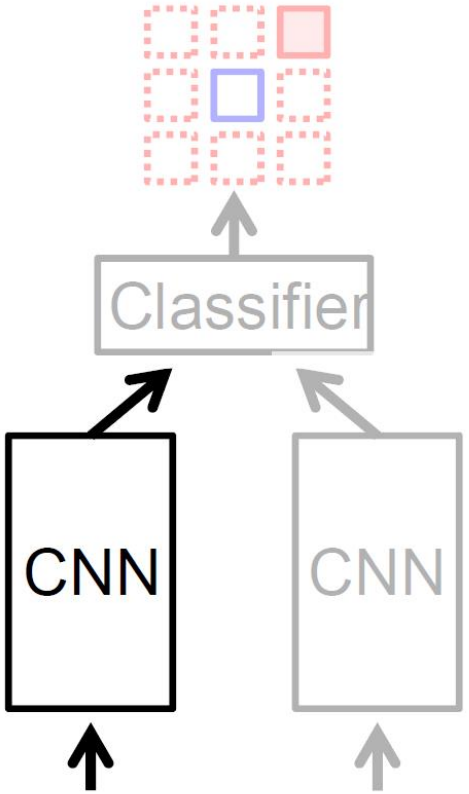


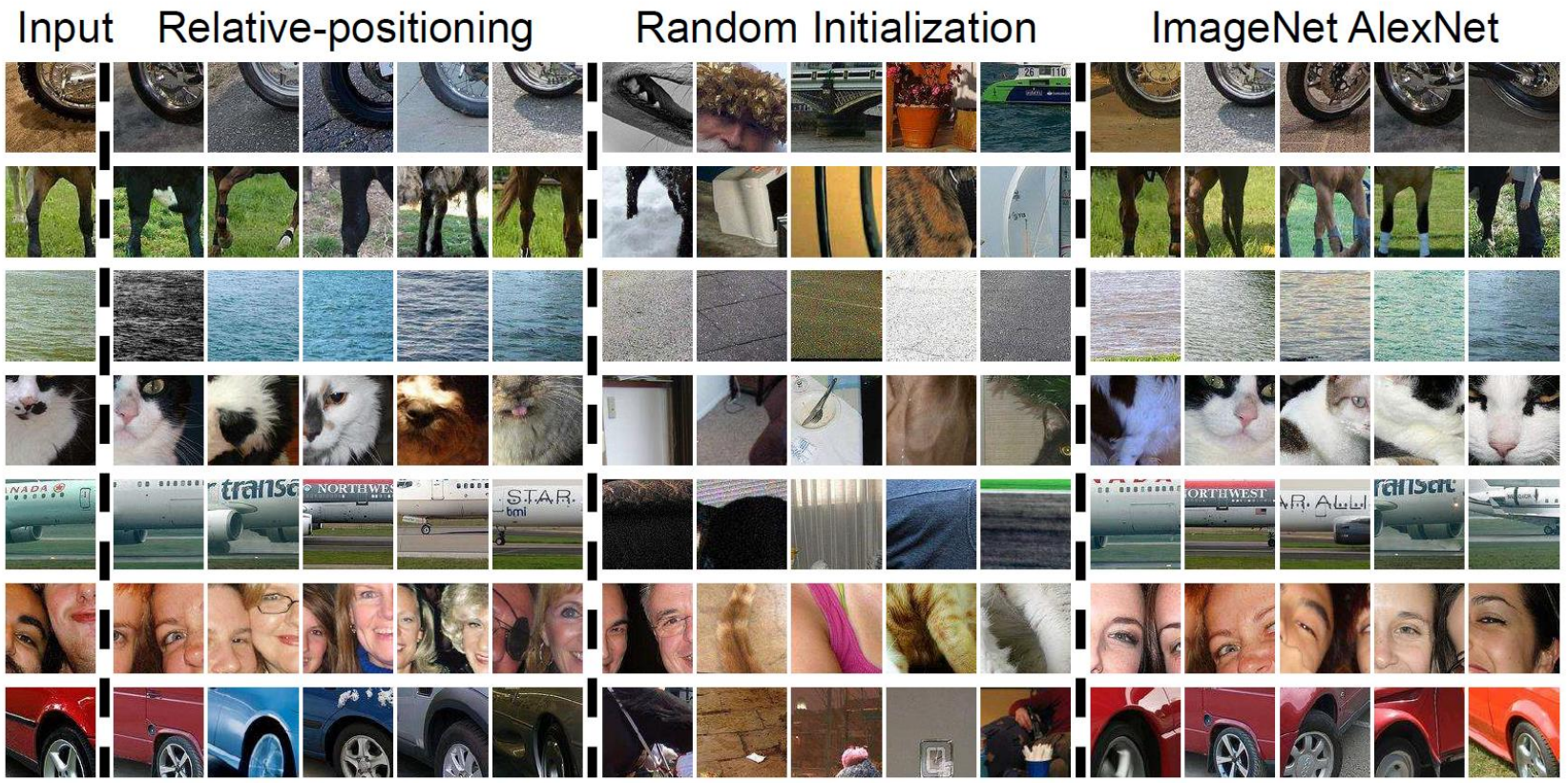
Randomly Sample Patch

Sample Second Patch

Unsupervised visual representation learning by context prediction,
Carl Doersch, Abhinav Gupta, Alexei A. Efros, ICCV 2015

What is learned?

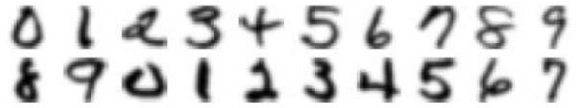




Semi-supervised Learning

Semi-supervised learning







Unlabeled data
Cheap and abundant !



Human expert/
Special equipment/
Experiment

"Crystal" "Needle" "Empty"

"0" "1" "2" ...

"Sports" "News" "Science" ...

Labels
Expensive and scarce !

Semi-Supervised learning: example 1

How unlabeled data helps?

Decision Boundary

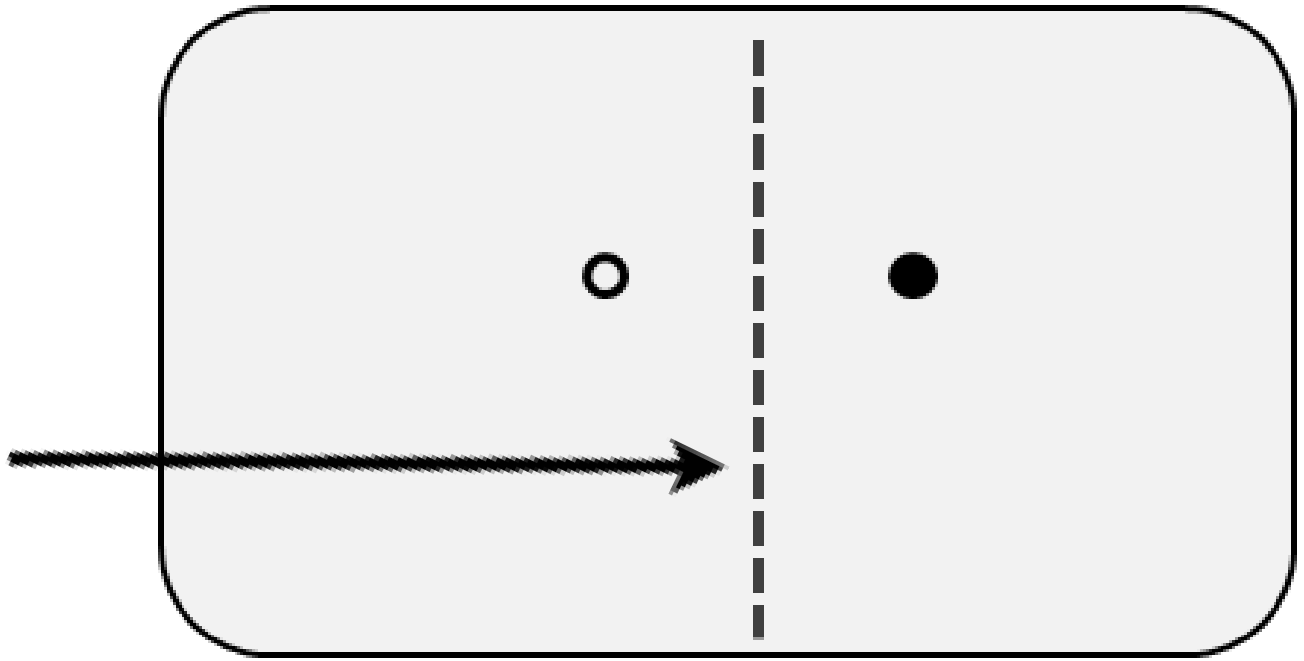
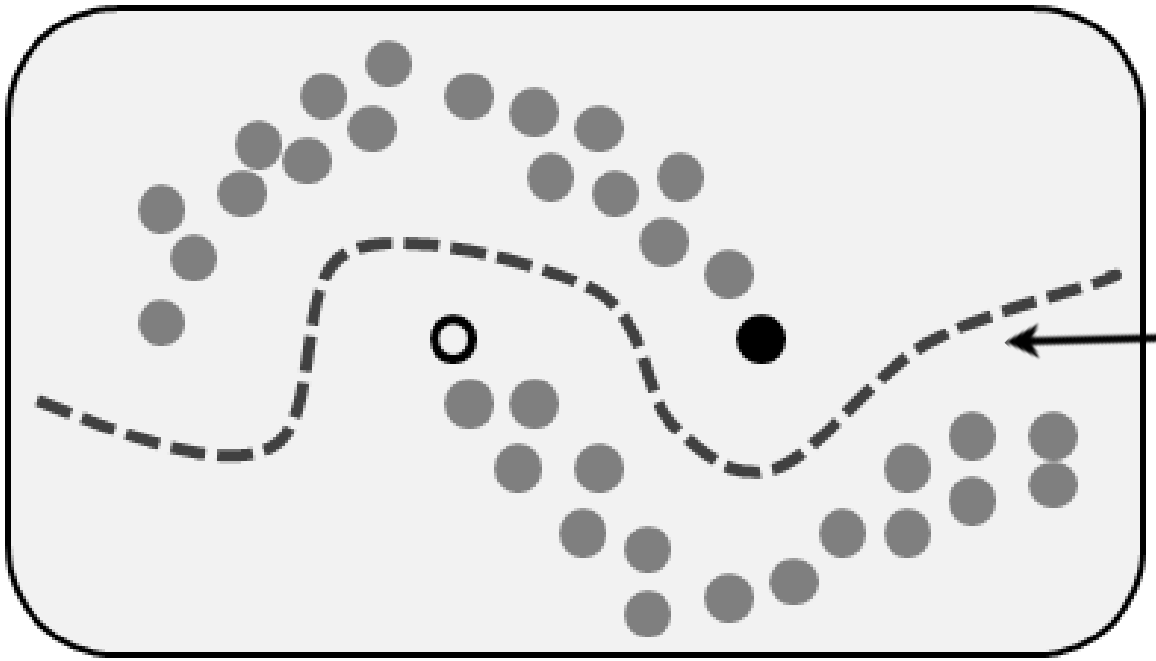


Image via “https://en.wikipedia.org/wiki/Semi-supervised_learning”

Semi-Supervised learning: example 1

Unlabeled data help a lot

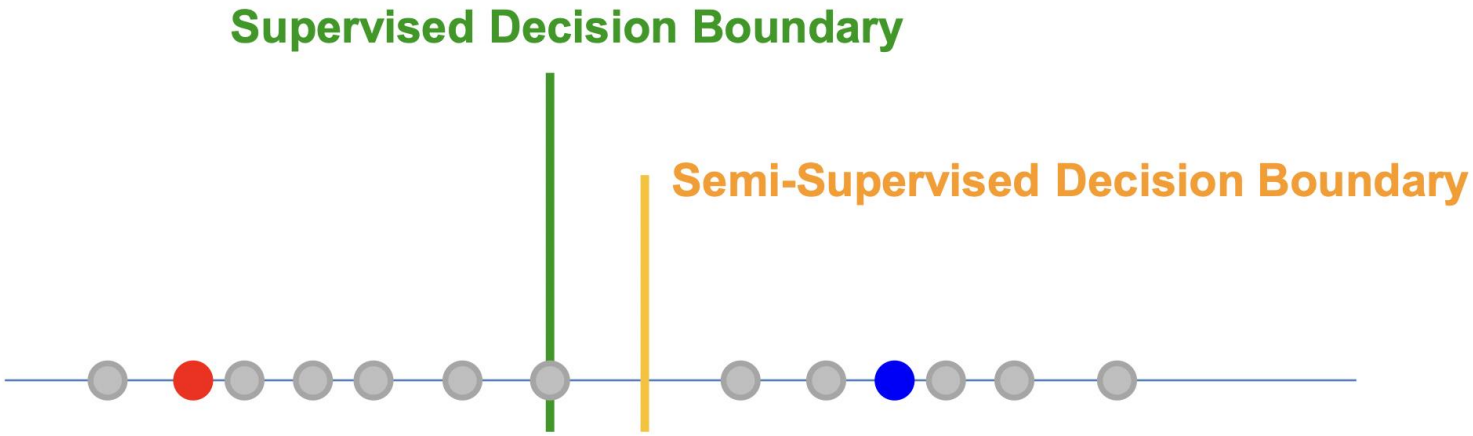


Better Decision Boundary

Image via “https://en.wikipedia.org/wiki/Semi-supervised_learning”

Semi-Supervised learning: example 2

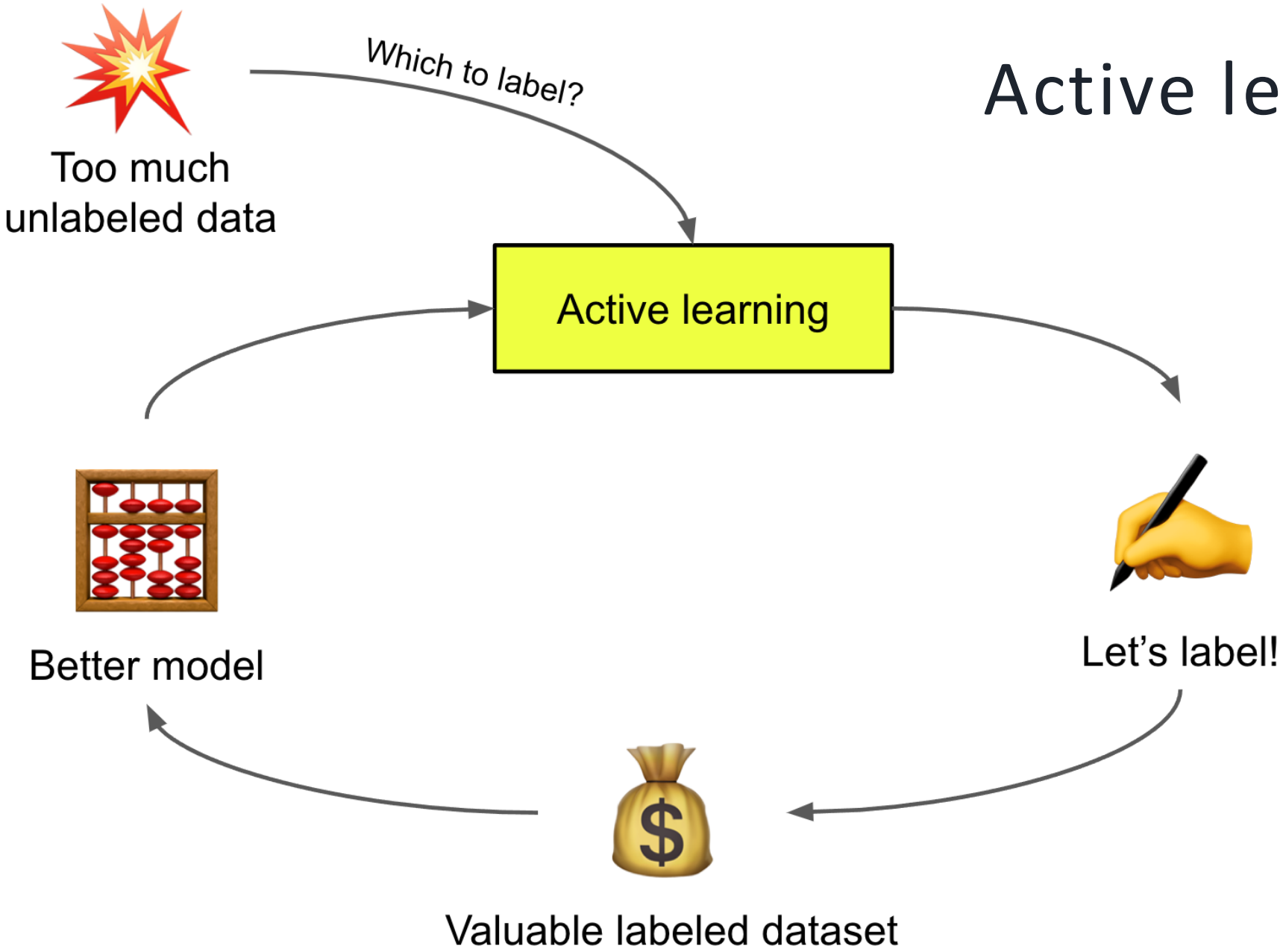
- Positive labeled data
- Negative labeled data
- Unlabeled data



Assume each class is a coherent group
Then unlabelled data can help identify the boundary more accurately.

Active Learning

Active learning



Active learning

- **Better data is often more useful than simply more data (quality over quantity)**

- **Data collection may be expensive**
  - ➢ Cost of time and materials for an experiment
  - ➢ Cheap vs. expensive data
    - Raw images vs. annotated images

- **Want to collect best data at minimal cost**

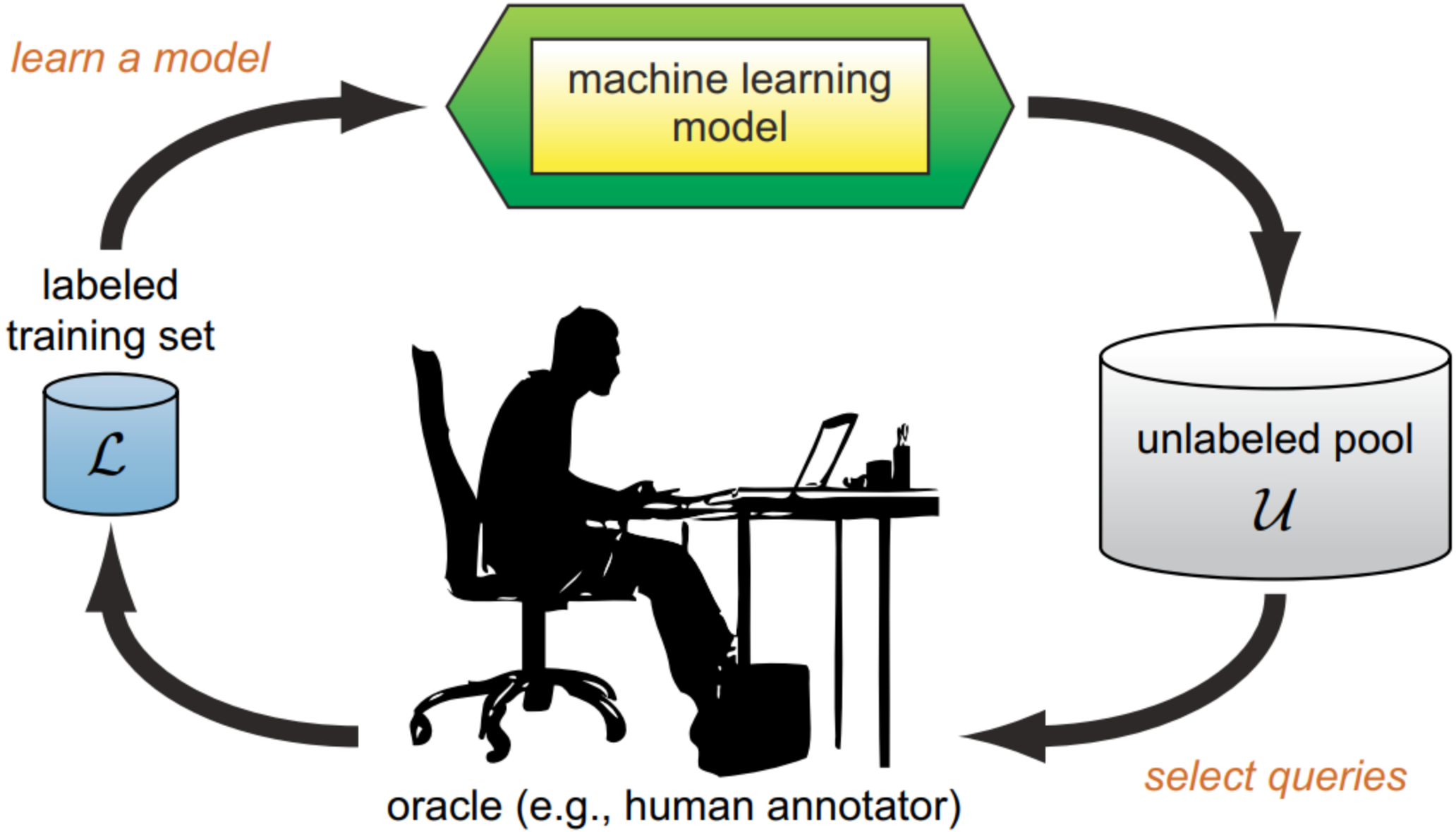


Figure 1: The pool-based active learning cycle.

Transfer Learning

Transfer Learning

- **Why do we care about transfer learning?**



(image: kuailexiaorongrong.blog.163.com, via https://sg.news.yahoo.com/6-kungfu-moves-movies-wished-194241611.html)

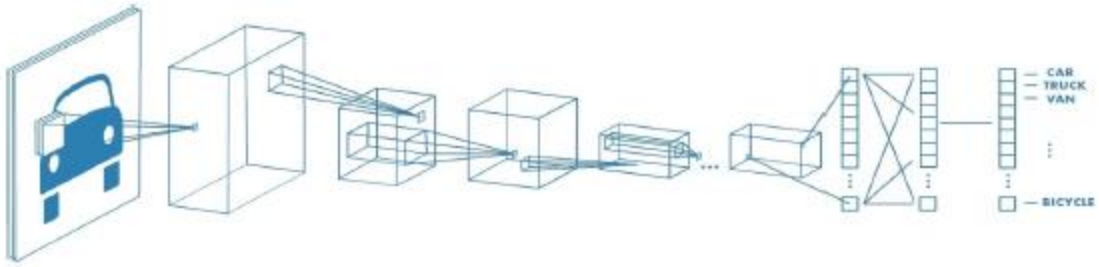
The recipe in vision and language domain

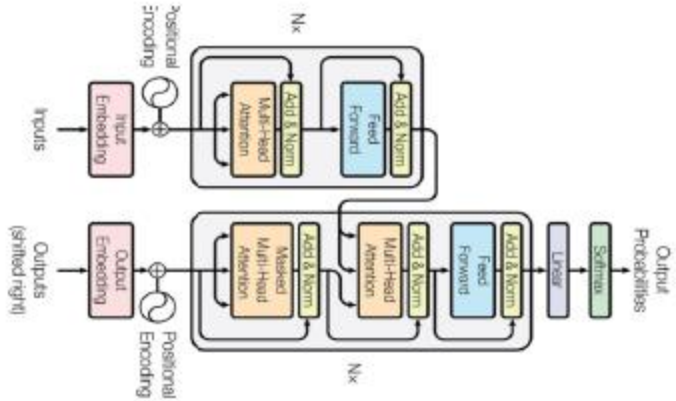
A lot of data





Expressive, capable models





What if you want a more general-purpose AI system?

Learning each task from scratch won't cut it.

What if you don't have a large dataset?

medical imaging

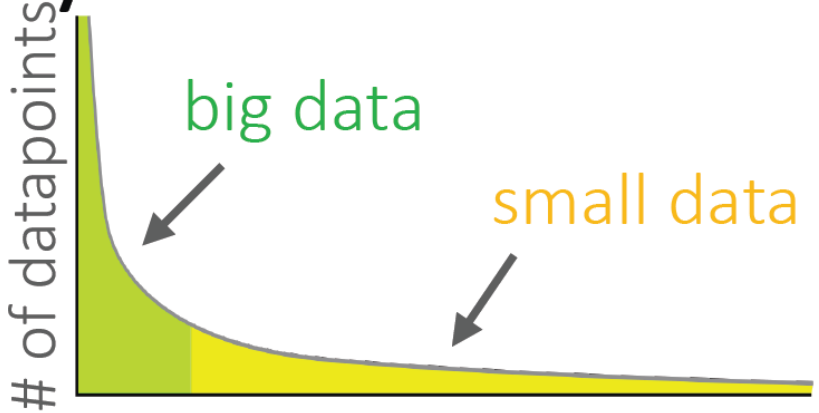
robotics

personalized education,

translation for rare languages

medicine, recommendations

What if your data has a long tail?

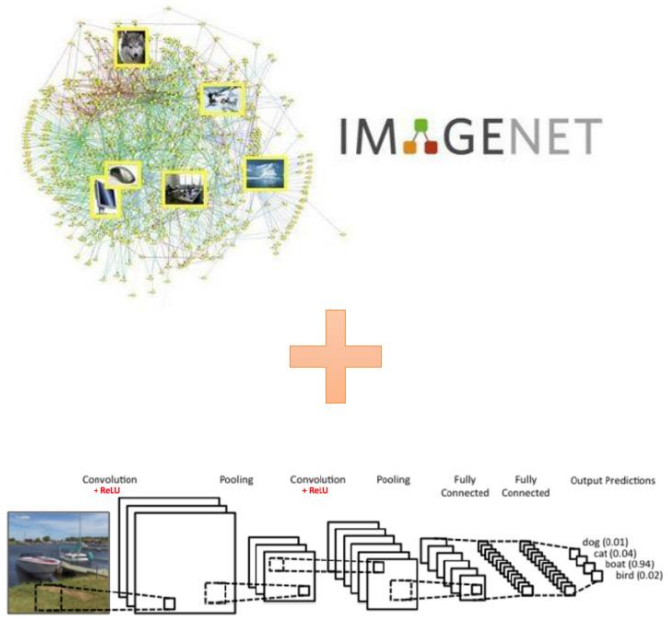


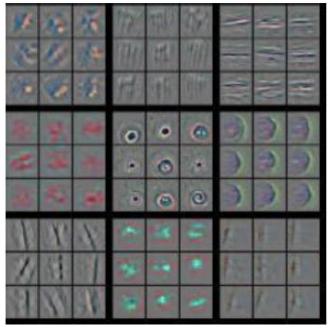
What if you need to quickly learn something new?

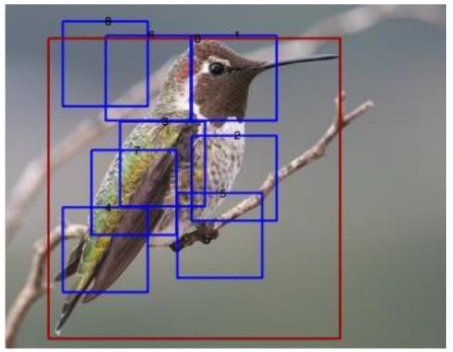
about a new person, for a new task, about a new environment, etc.

Fine-tuning

The most popular transfer learning method in (supervised) deep learning!

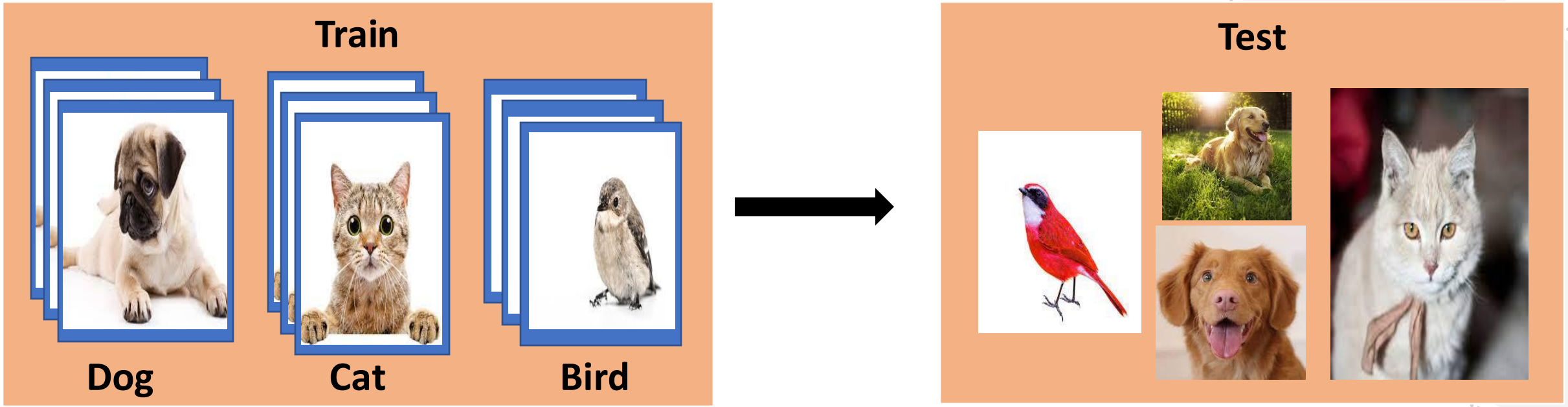


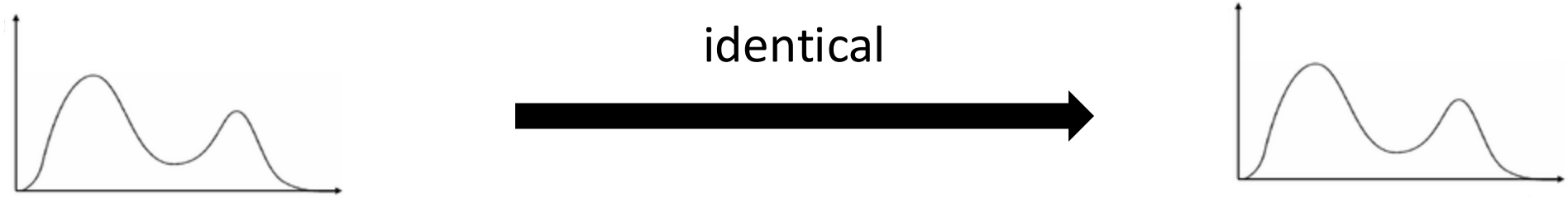




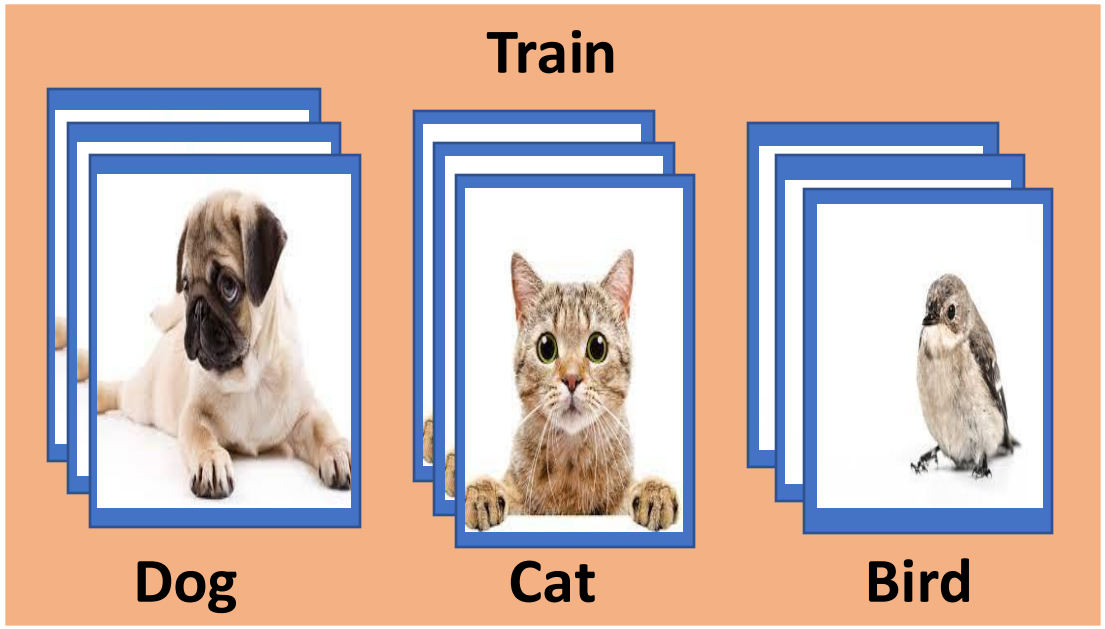
Domain Adaptation & Generalization

An over-simplified assumption: independent and identically distributed

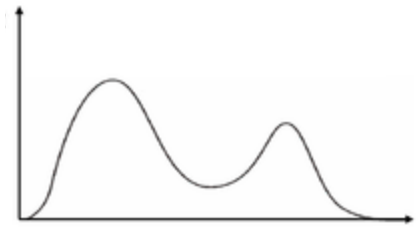




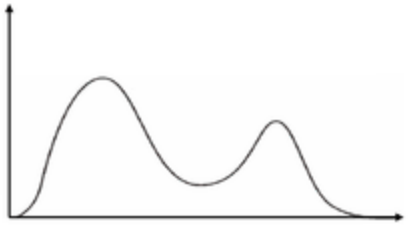
An over-simplified assumption: independent and identically distributed



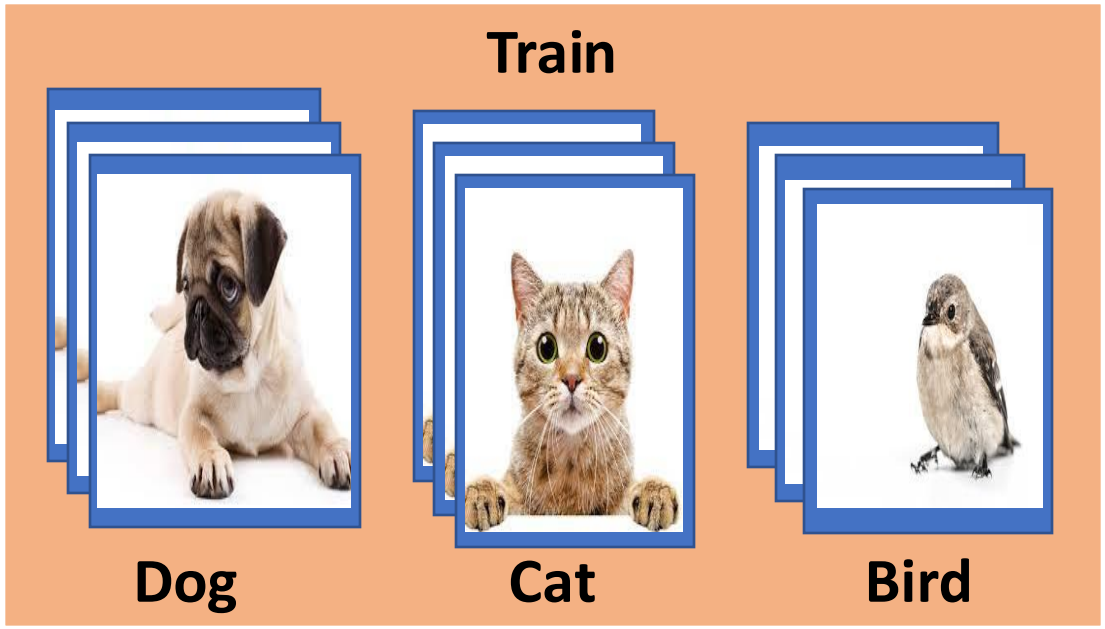


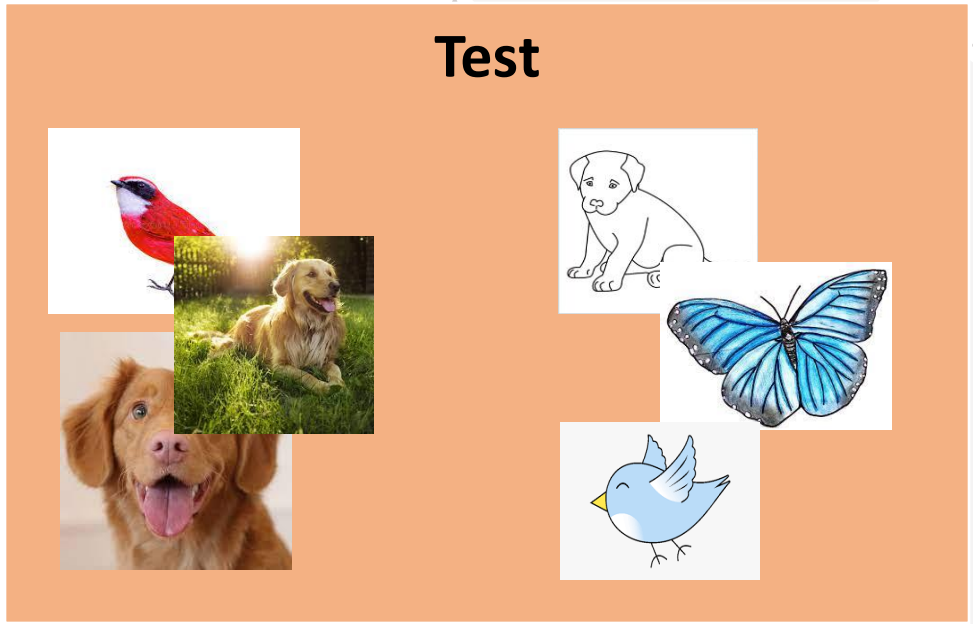


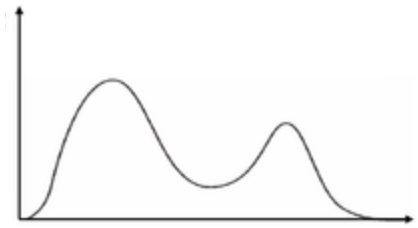
identical



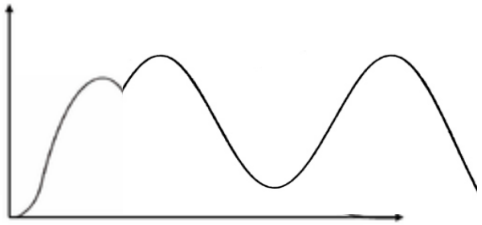
A more realistic scenario: distribution shifts



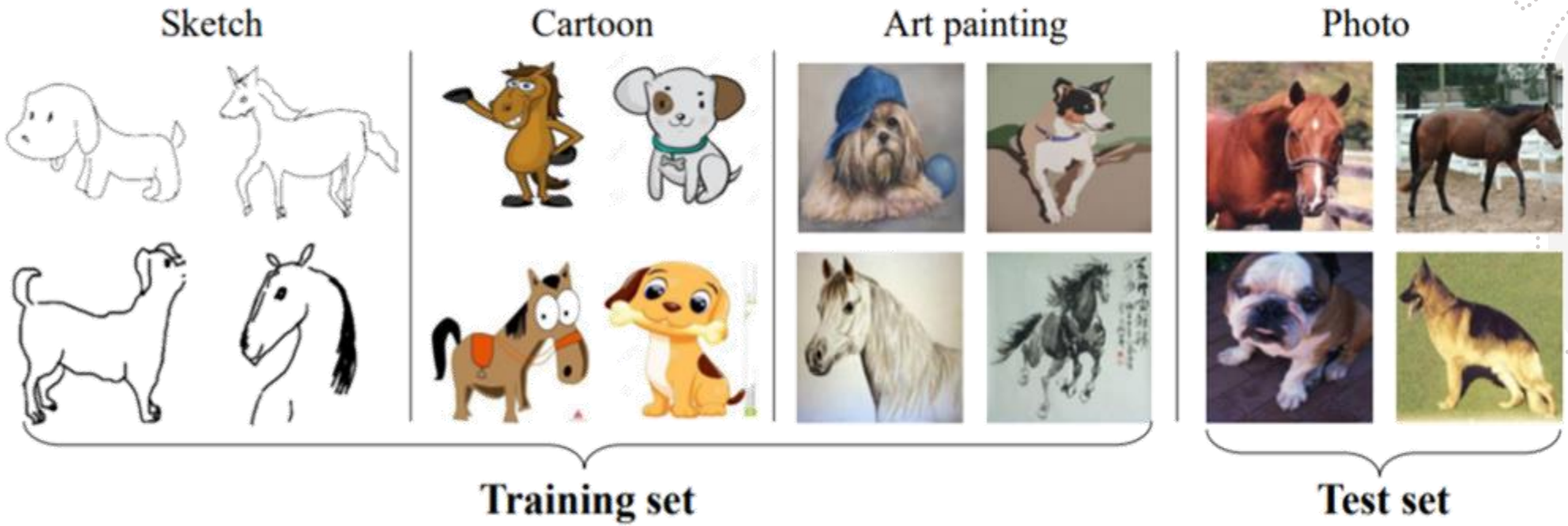




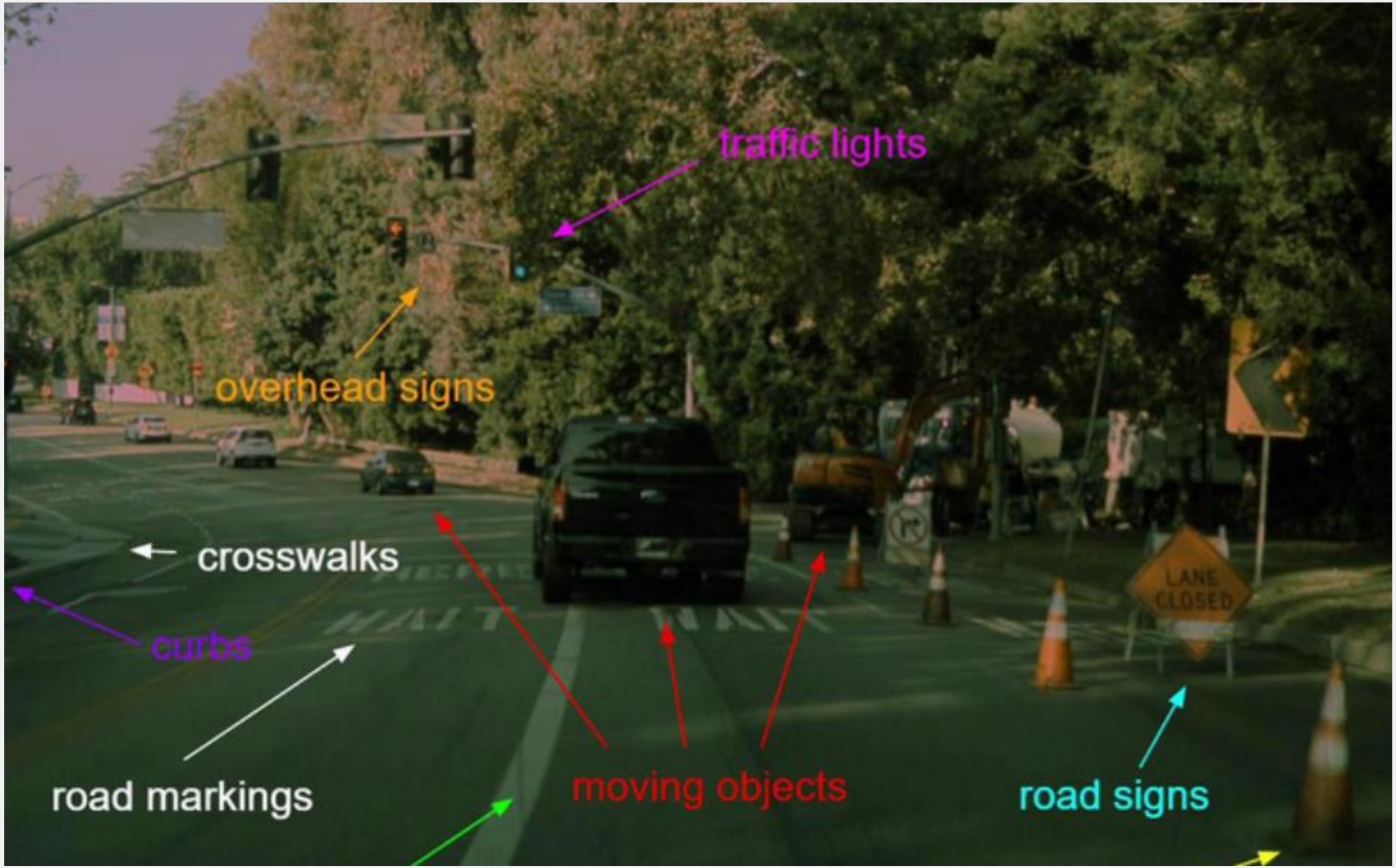
identical



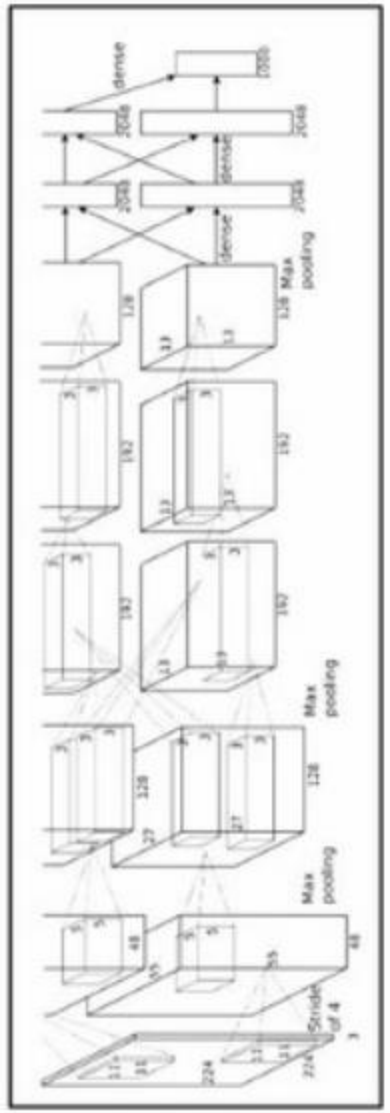
What makes domain generalization hard?



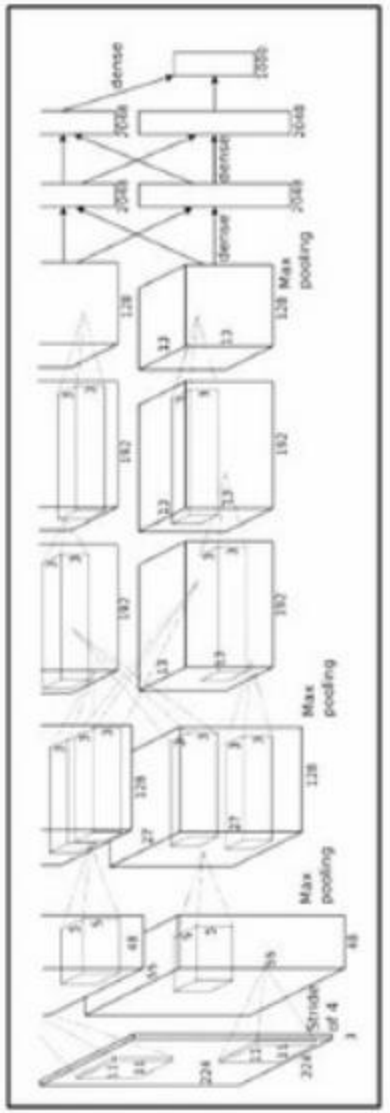
Multi-task Learning


traffic lights
overhead signs
crosswalks
curbs
road markings
moving objects
road signs
LANE CLOSED

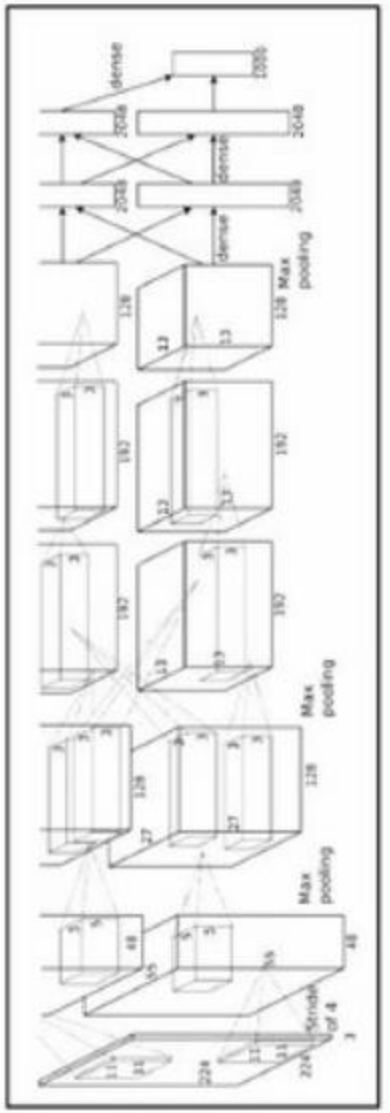
moving objects



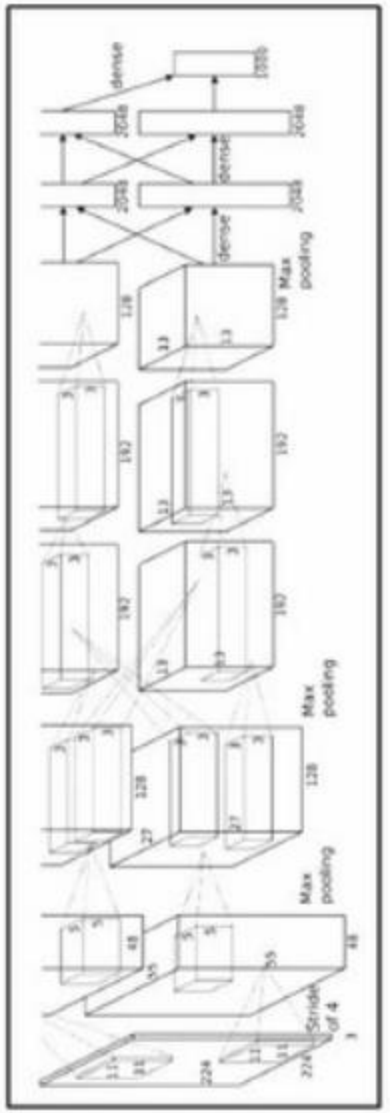
static objects



signs



traffic lights



What is the critical assumption?

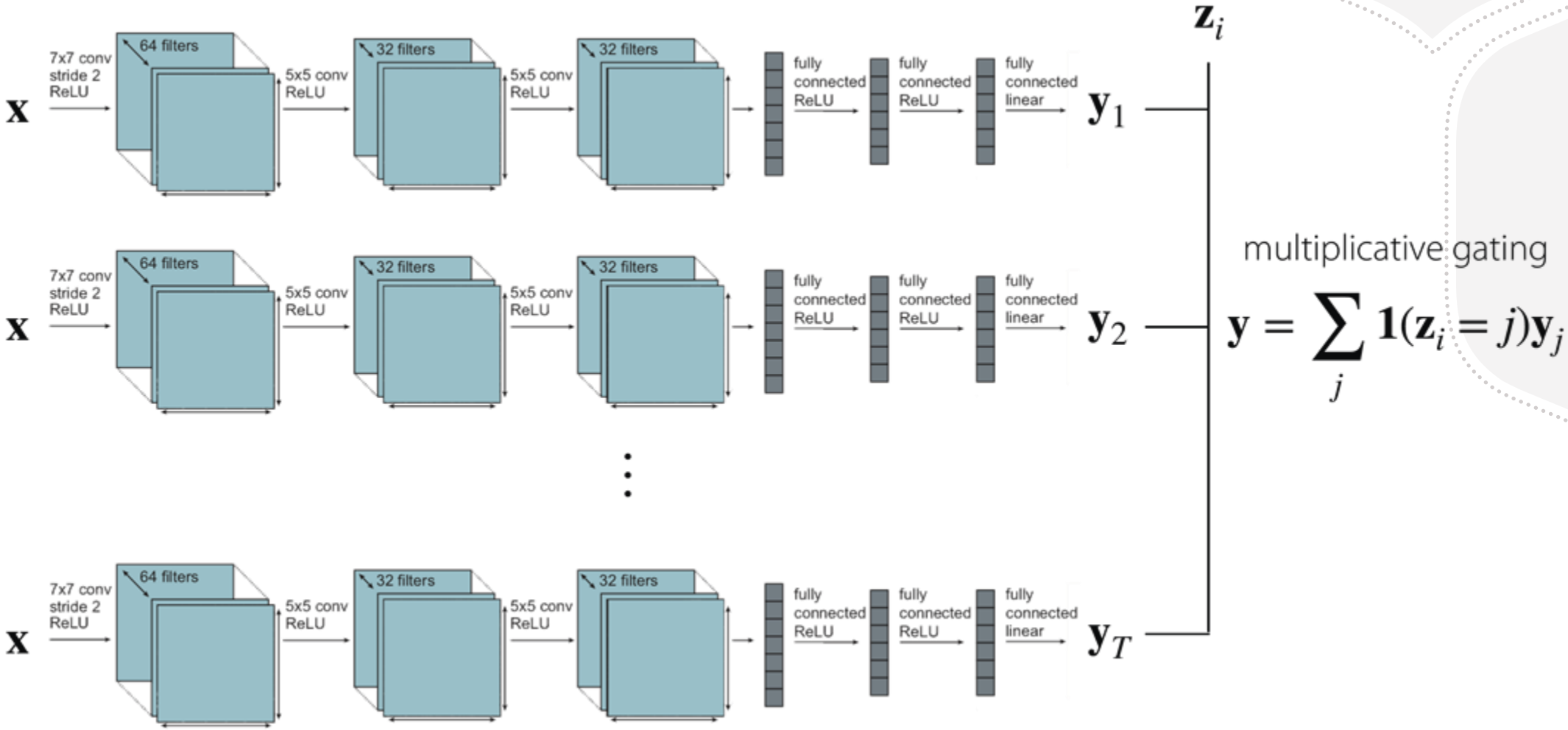






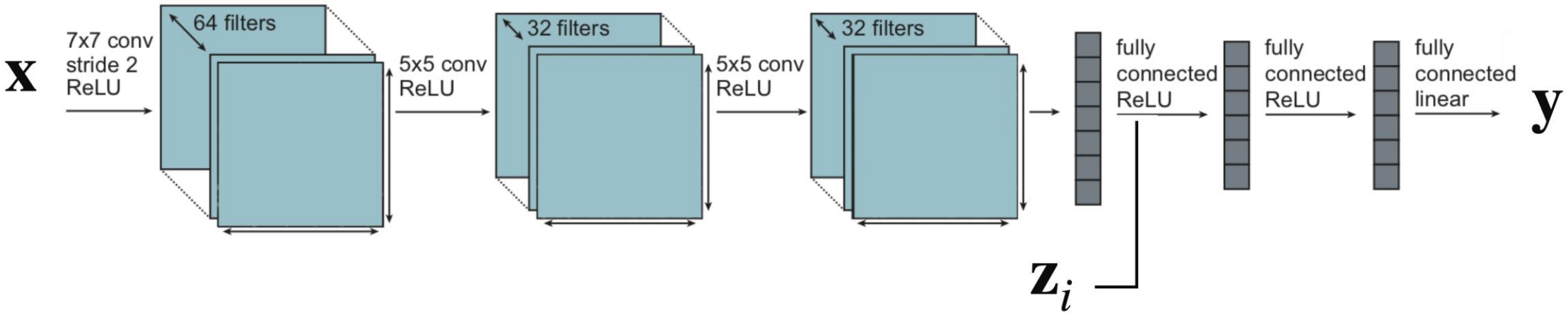
Conditioning on the task

- **Question: How should you condition on the task in order to share as little as possible?**

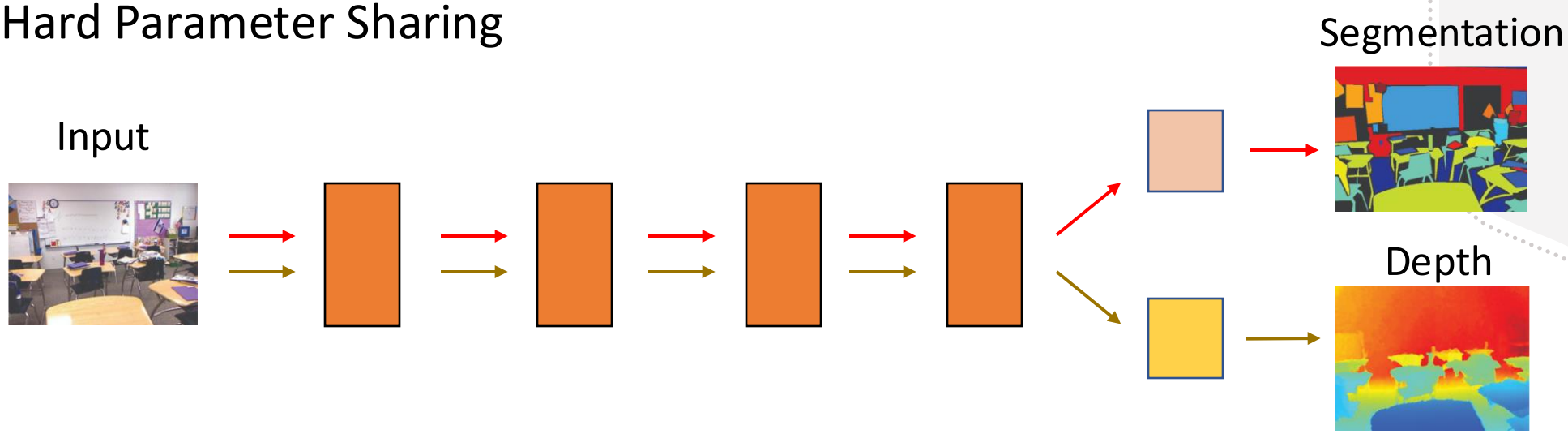


Conditioning on the task

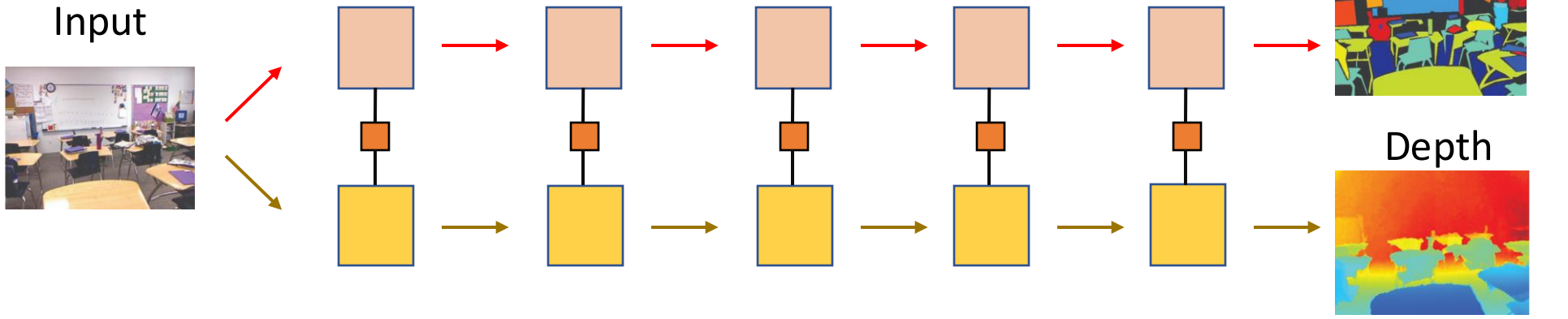
- **Question: How should you condition on the task in order to share as much as possible?**

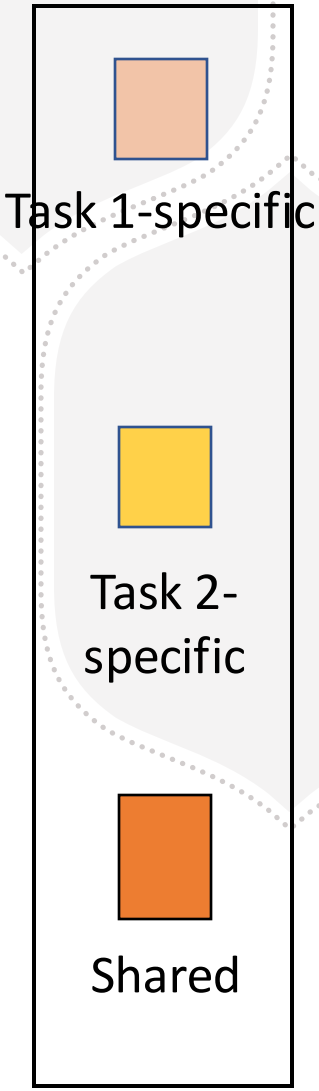


Hard Parameter Sharing



Soft Parameter Sharing





Challenges

- Challenge #1: Negative transfer
- Challenge #2: Overfitting
- Challenge #3: What if you have a lot of tasks?

Meta Learning

Goal

Machine Learning ≈ find a function f

Dog-Cat Classification $f($  $) =$ 'Dog'

Meta Learning ≈ find a function F **that finds a function f**

Meta Learning

Using image classification as the example, meta learning learns a **general rule for distinguishing classes** that transfers across different classification tasks

Traditional Supervised Learning

- **Classification Task:** classify 5 species of birds.
- **Limitation:** How to classify 5 types of phones?

Meta Learning

- **Task 1**: classify 5 species of birds (5-way, 1-shot)
- **Task 2**: classify 5 types of flowers
- **Task 3**: classify 5 types dog
- **Task 4**: classify 5 car models
- …
- **Target:** classify 5 types of phones

	Traditional Classification	Meta-Learning
Training data	One dataset, many examples per class	Many tasks, few examples per class
Goal	Solve one task well	Quickly solve *new* tasks
Output layer	Fixed to training classes	Flexible — adapts to new classes
Test time	Same classes as training	**New, unseen classes**
Data needed for new task	Thousands of labels	A handful (1–5 per class)
What's learned	Class-specific features	A prior / initialization / metric that generalizes across tasks

Example of Meta Learning



Given 1 example of 5 classes:

Classify new examples



Can replace image classification with: regression, language generation, skill learning, **any ML problem**

The Meta-Learning Problem

Given data from tasks: $\{\mathcal{T}_i\}_{i=1}^{n}$ Quickly solve new task: $\mathcal{T}_{test}$

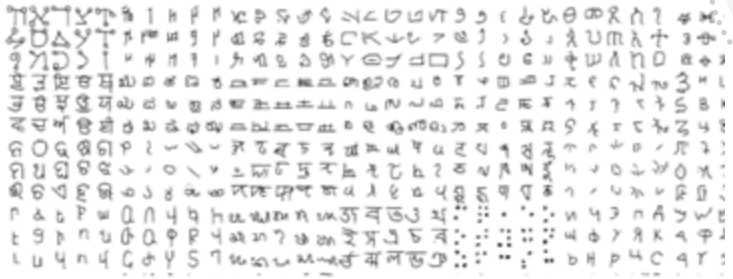
Key assumption: meta-training tasks and meta-test task drawn i.i.d. from same task distribution

Optimization Goal: $\theta = \underset{\theta}{argmin}\, \mathbb{E}_{\mathcal{T}\sim p(\mathcal{T})}[\mathcal{L}_\theta(\mathcal{T})]$

What do the tasks correspond to?

- recognizing handwritten digits from different languages (
- giving feedback to students on different exams
- classifying species in different regions of the world
- a robot performing different tasks





How many tasks do you need? The more the better. (analogous to more data in ML)

Problem Settings

Multi-Task Learning

Solve multiple tasks $\mathcal{T}_1, \cdots, \mathcal{T}_T$ at once.

$$\min_{\theta} \sum_{i=1}^{T} \mathcal{L}_i(\theta, \mathcal{D}_i)$$

The Meta-Learning Problem

Given data from $\mathcal{T}_1, \ldots, \mathcal{T}_n$, quickly solve new task $\mathcal{T}_{\text{test}}$

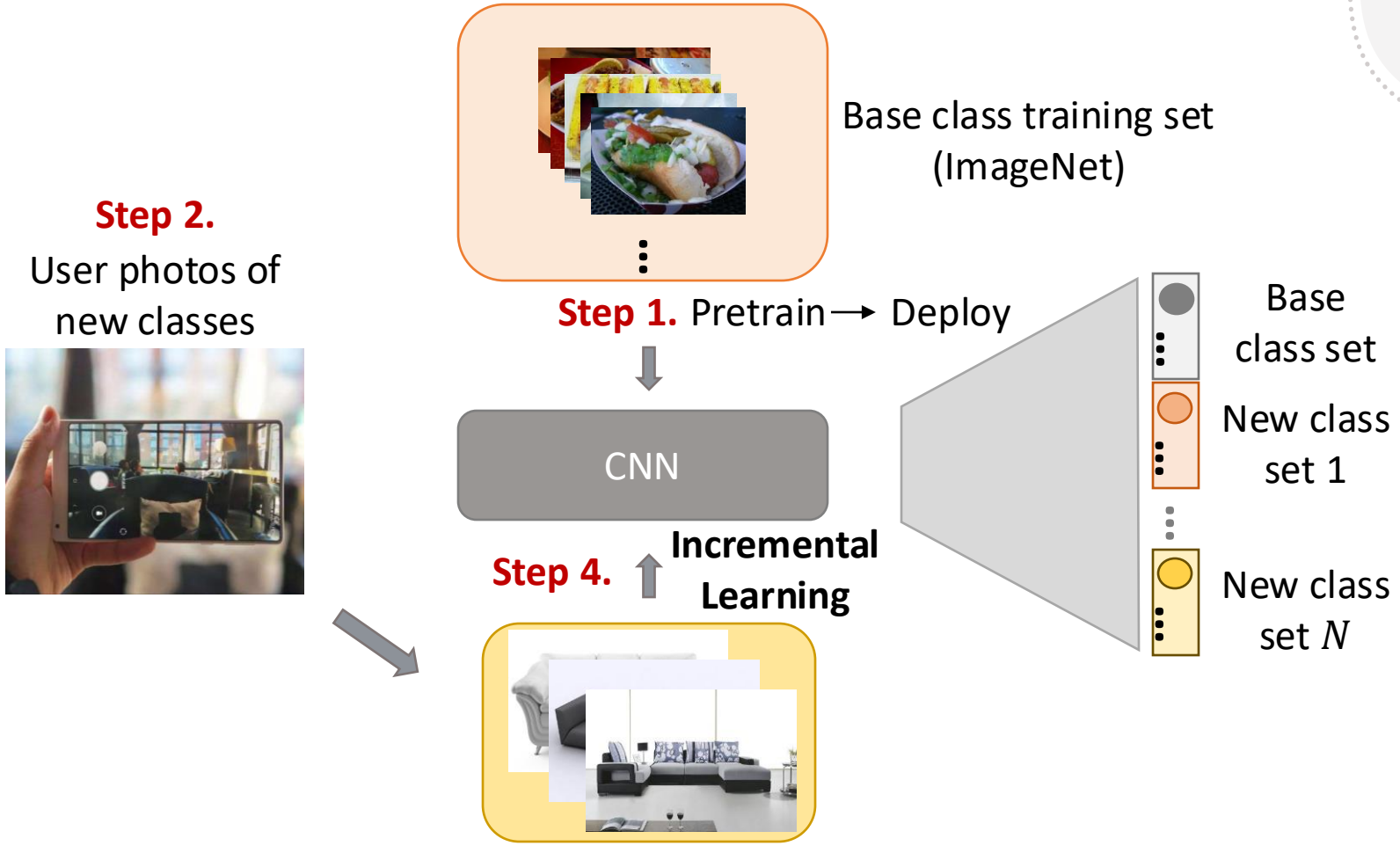
Transfer Learning

Solve target task $\mathcal{T}_b$ after solving source task $\mathcal{T}_a$ by *transferring* knowledge learned from $\mathcal{T}_a$

In all settings: tasks must share structure.

Incremental Learning

What is class-incremental learning and why?



Class-incremental learning: conditions & challenges

Goal: **Incrementally** learn **new** classes

Session 0



10 classes, 600 samples per classes

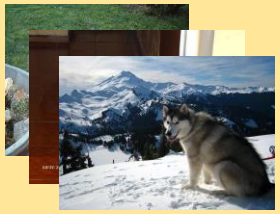
Session 1



10 classes, 600 samples per classes

...

Session 9



10 classes, 600 samples per classes

Class-incremental Learner

10-classes classifier → 20-classes classifier → ... → 100-classes classifier

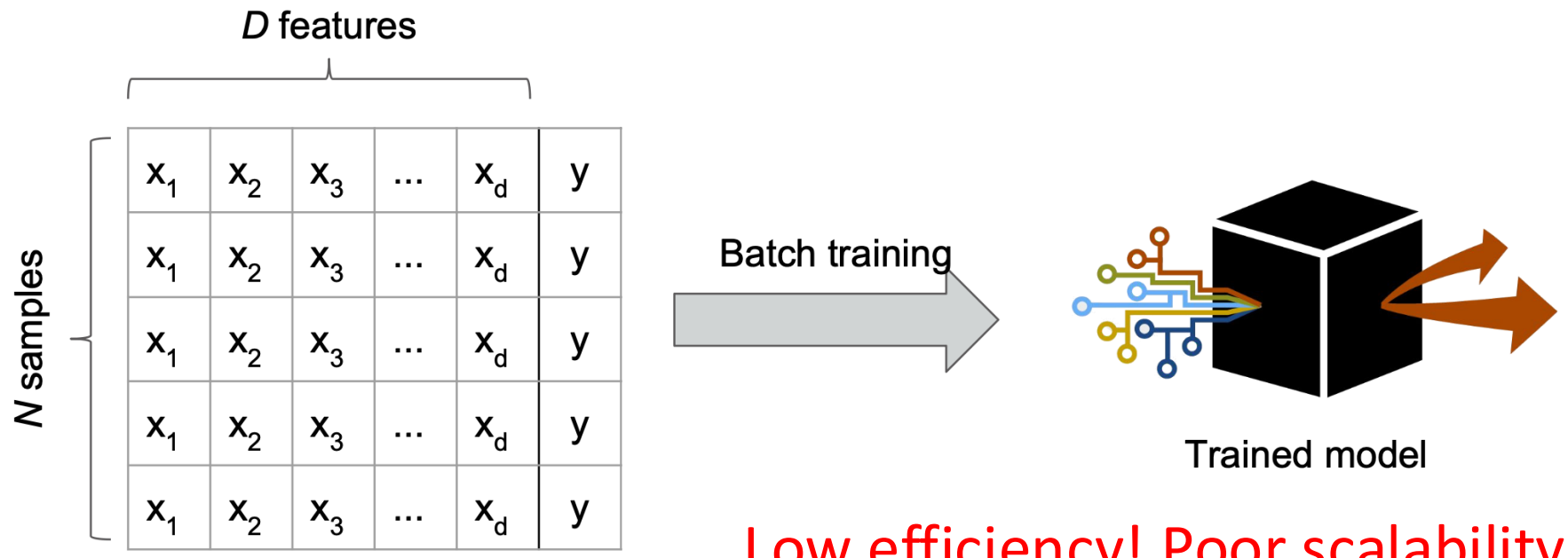
Challenges



Online Learning

Offline learning

- **In batch (or offline) learning you have access to the whole dataset to train on.**

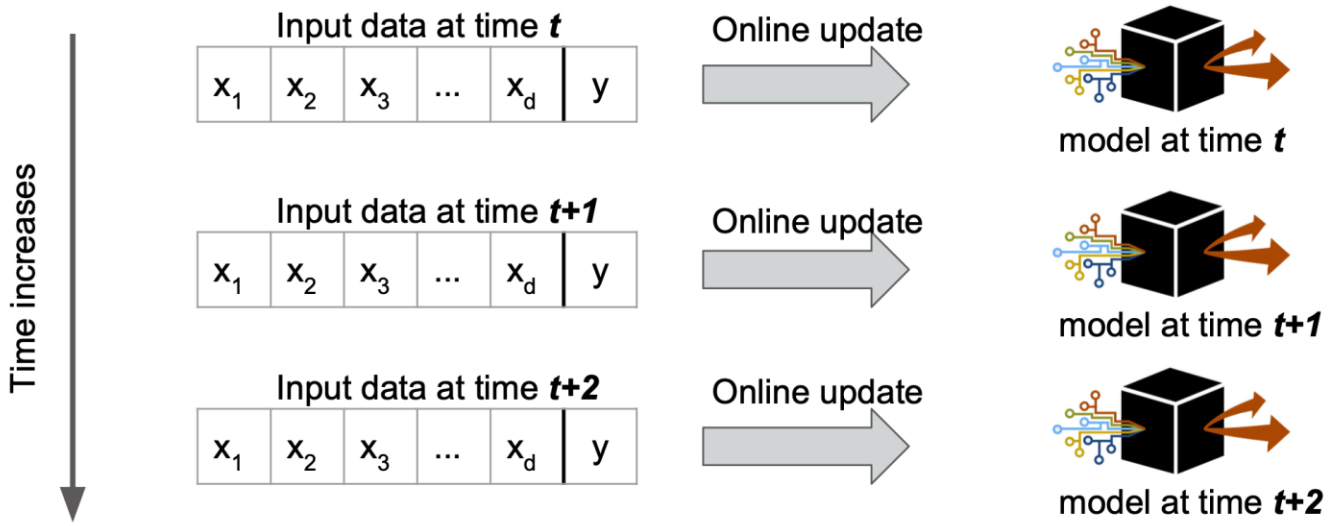


Low efficiency! Poor scalability!

Online learning

- **In online learning, on each round,**

  Receiving a data instance → Making a prediction → Receiving true answer from environment as a feedback → Updating model by some strategy to improve performance on future received instances



Overcoming the drawbacks of batch learning in that the predictive model can **be updated instantly for any new data instances**

Online learning

- **Online learning is generally described as doing machine learning in a *streaming* data setting, i.e. training a model in consecutive rounds**

  - At the beginning of each round the algorithm is presented with an input sample, and must perform a prediction

  - The algorithm verifies whether its prediction was correct or incorrect, and feeds this information back into the model, for subsequent rounds

Application

- **Some cases where data is constantly being generated and you need quick predictions:**

  - Real-time Recommendation
  - Fraud Detection
  - Spam detection
  - Portfolio Selection
  - Online ad placement

Incremental Learning vs Online Learning

Online Learning:

- Receiving instances one by one
- Maybe can see instances from former classes (tasks)

Incremental Learning:

- Receiving instances by tasks
- Former classes are **inaccessible**
- Focus on solving **catastrophic forgetting**

Ensemble Learning


What is ensemble learning and why?

Intuition

- **Majority vote:**

  Suppose we have 5 completely independent classifiers.

  - If accuracy is 70% for each
    - $(0.7)^5 + 5 \times (0.7)^4(0.3) + 10 \times (0.7)^3(0.3)^2$
    - **83.7% majority vote accuracy**
  - With 101 such classifiers
    - **99.9% majority vote accuracy**

Value of ensembles

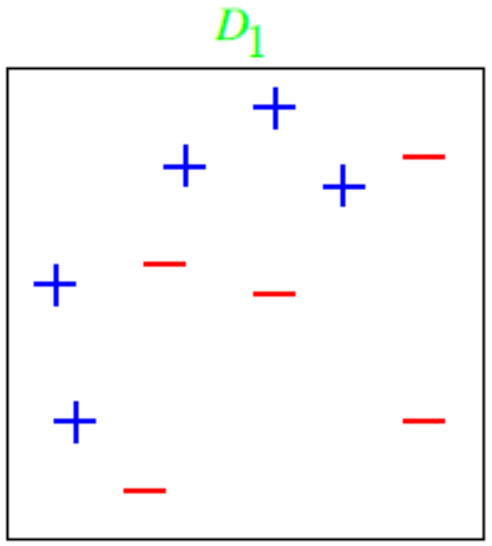
"NO FREE LUNCH" THEOREM



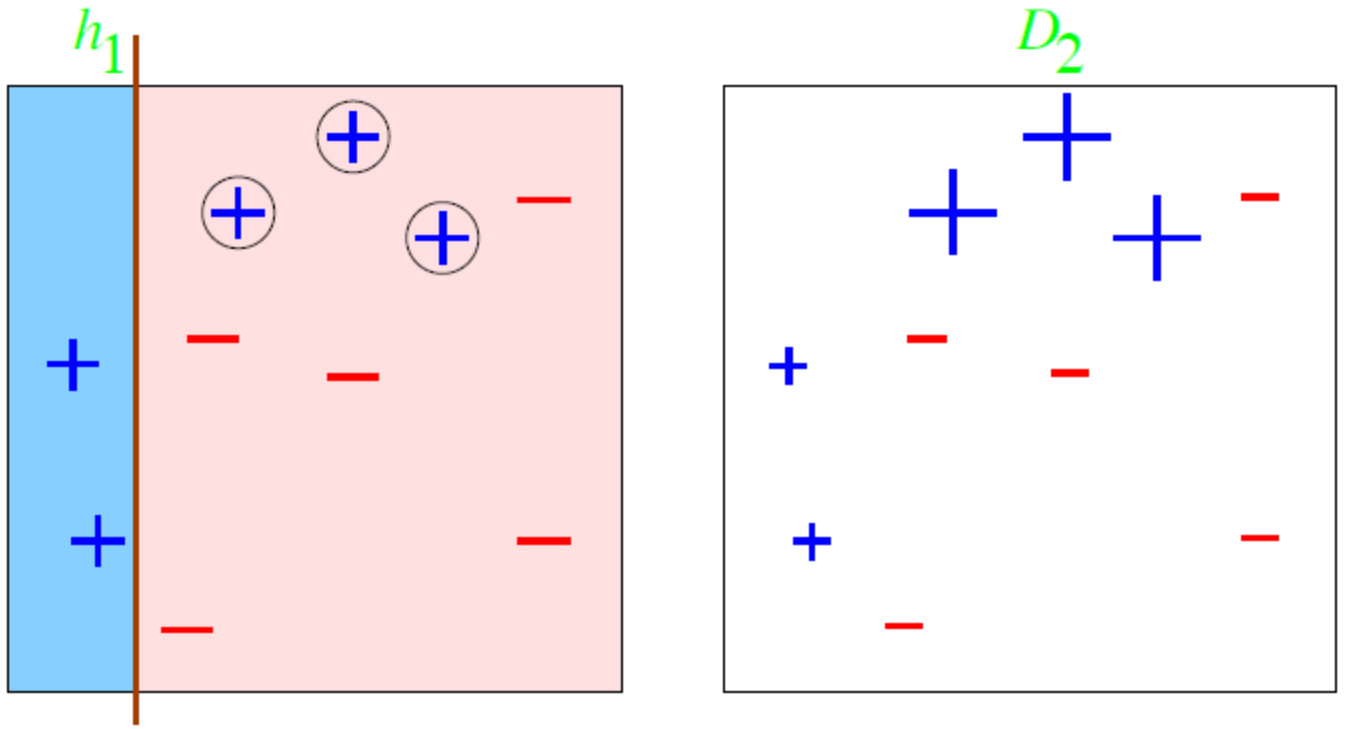
No single learning algorithm is best for all problems. Averaged over all possible problems, every algorithm performs equally well.

Example of AdaBoosting

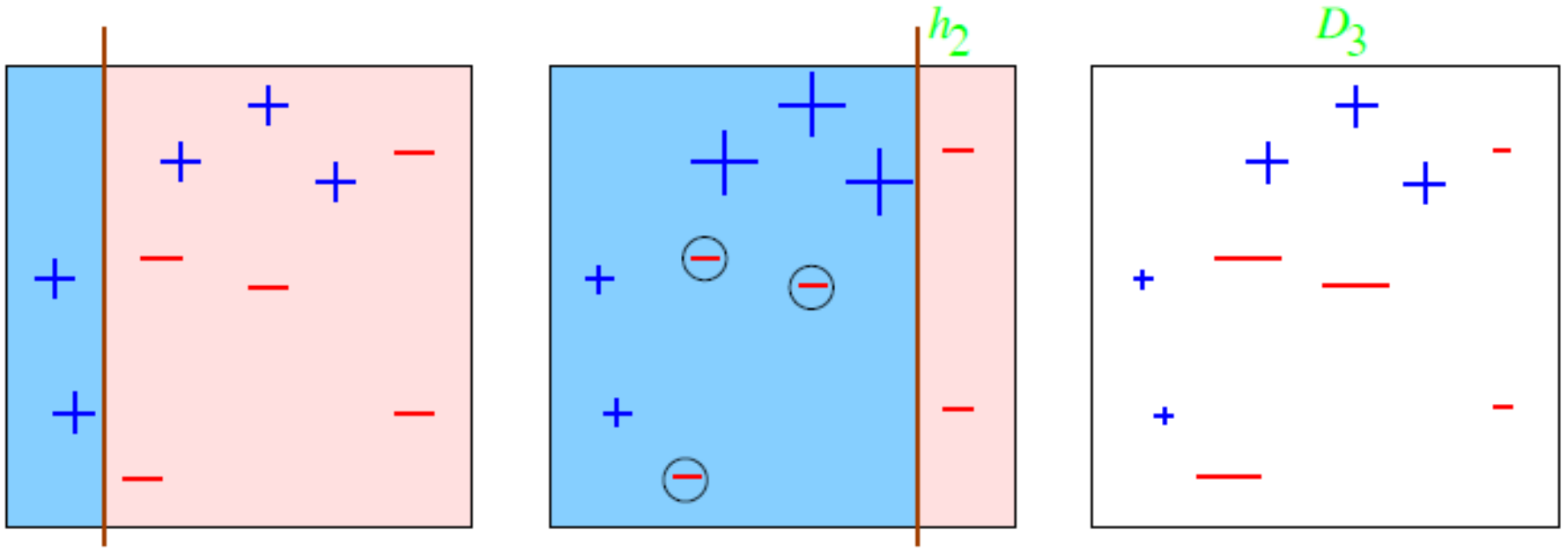
10 data points and 2 features



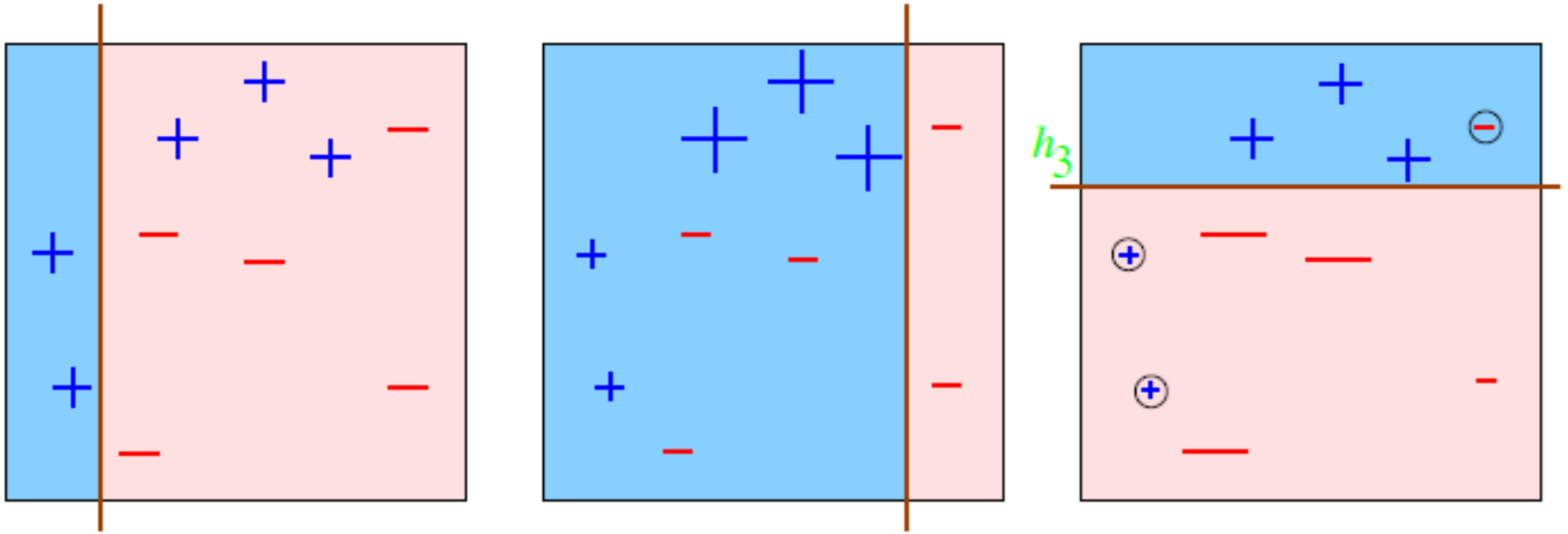
Round 1



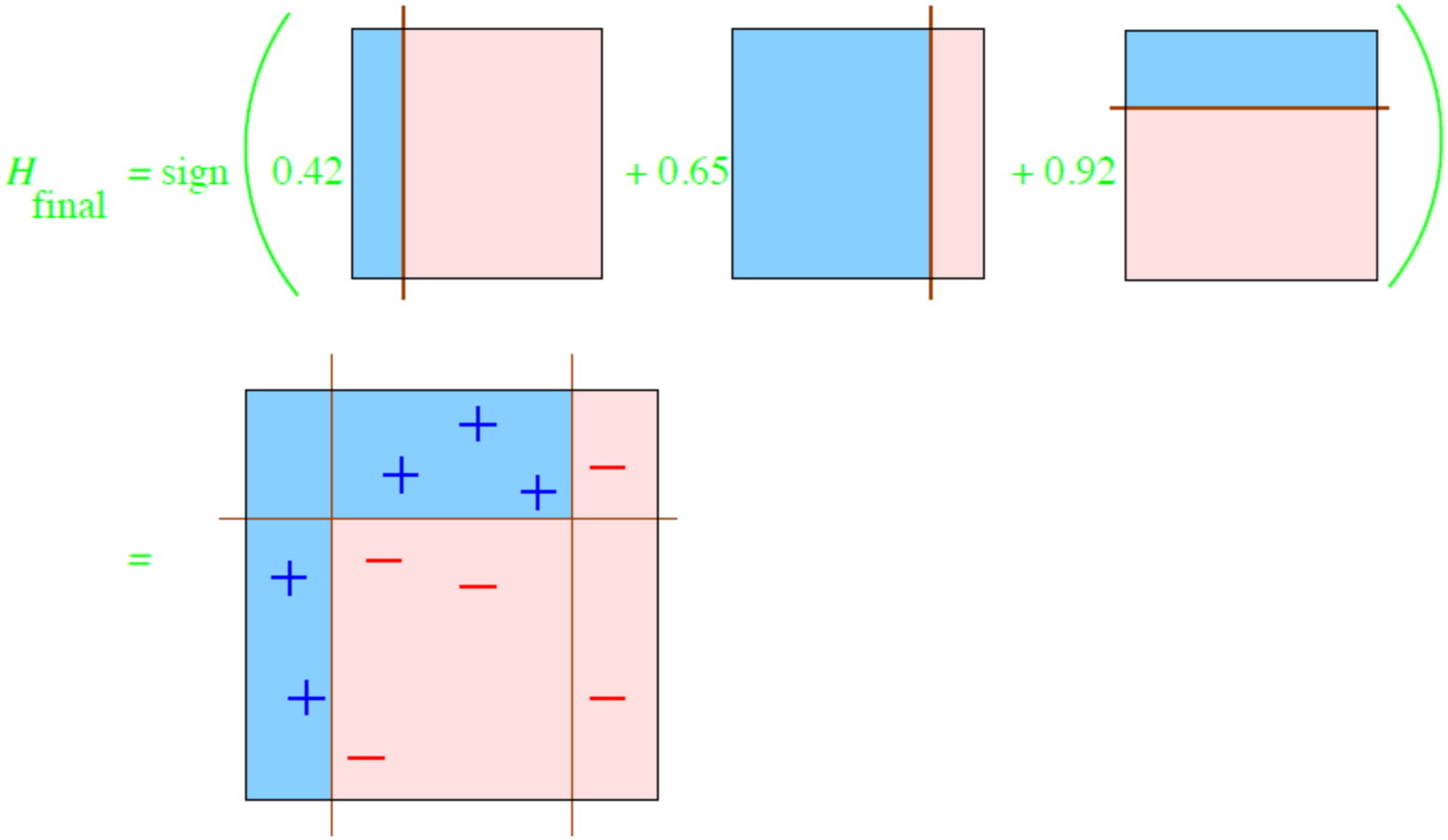
Round 2



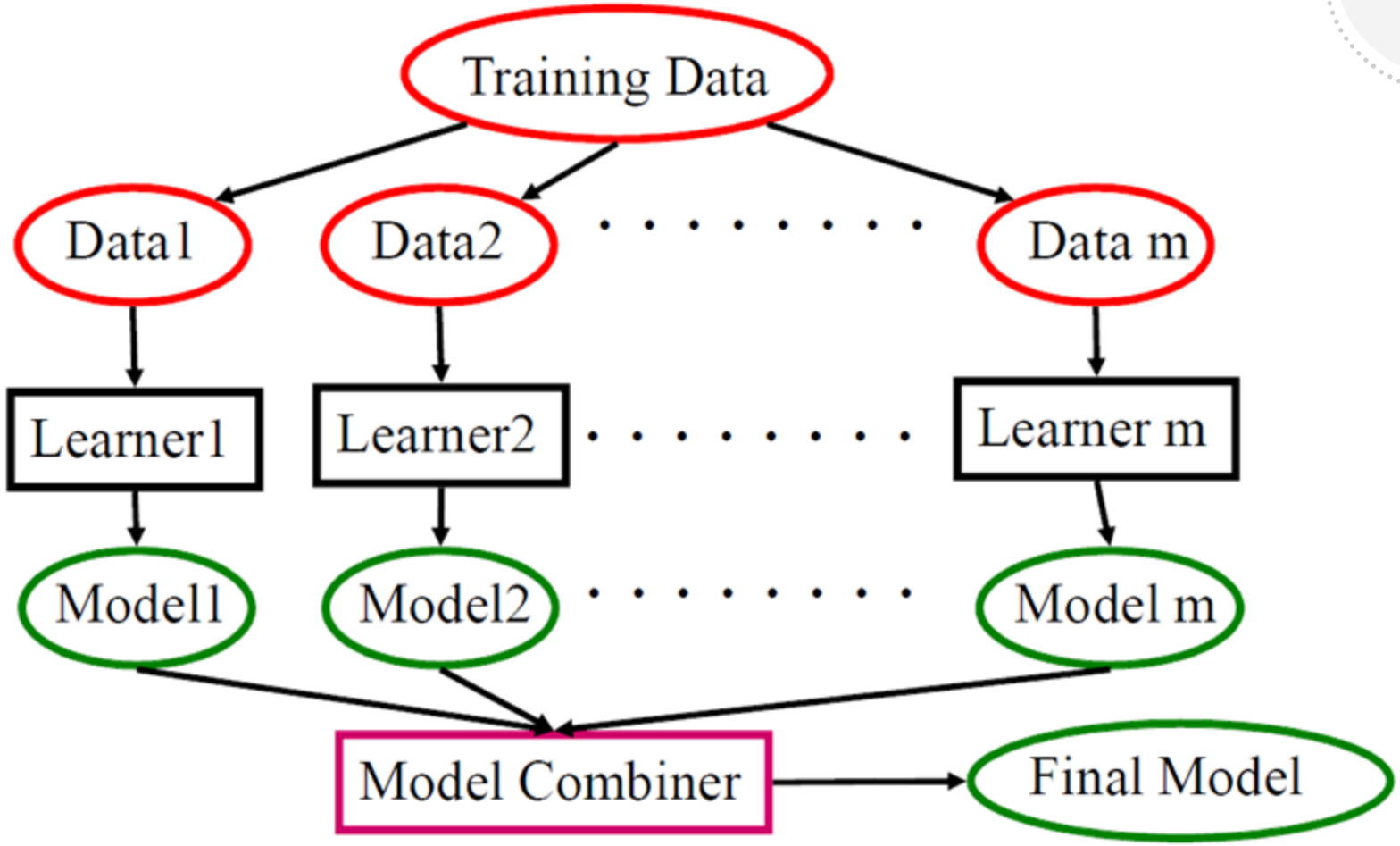
Round 3



Final classifier



Learning Ensembles

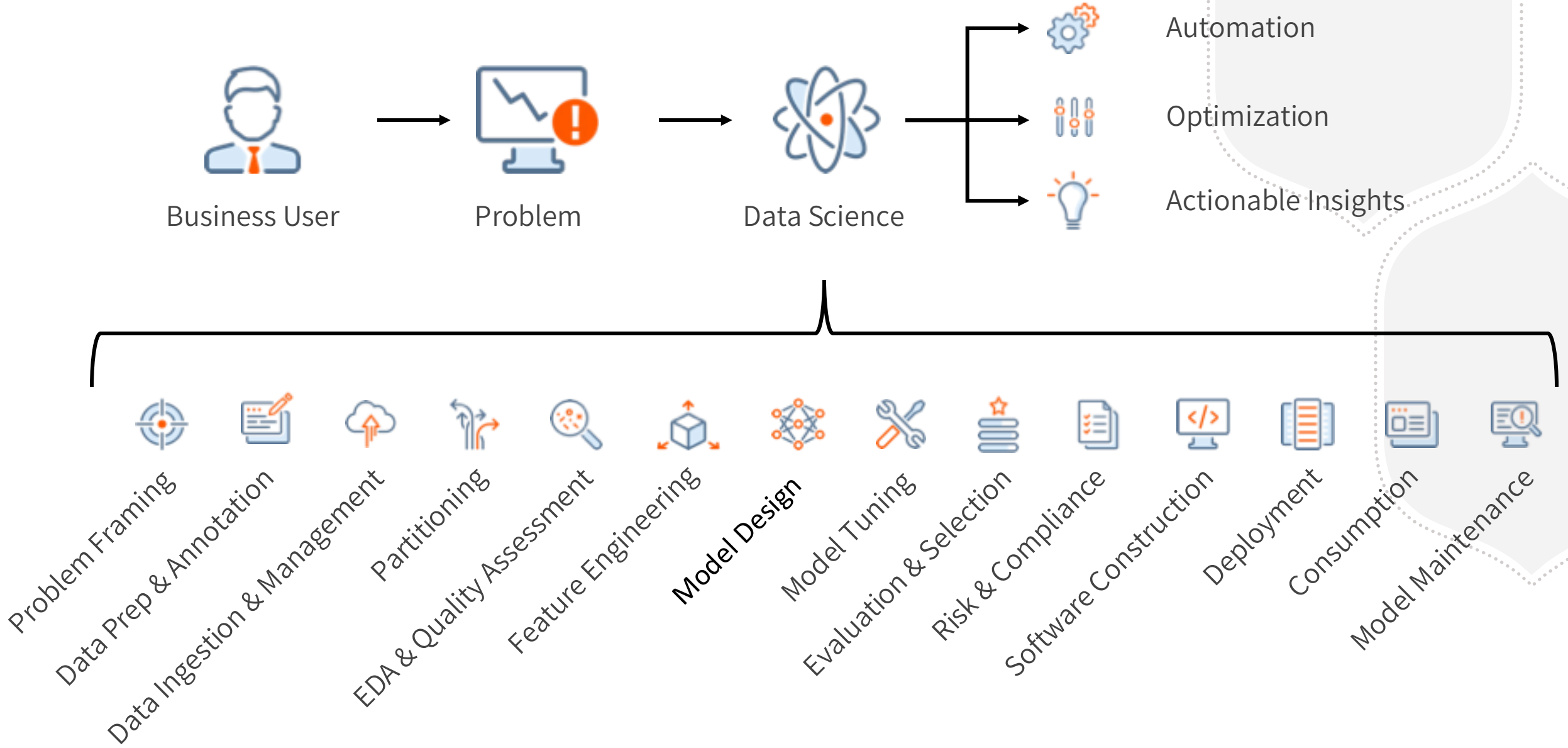


Question	Approaches	How It Works
How do we combine the outputs of base learners for maximum accuracy?	**Voting**	Aggregate predictions from multiple base learners — majority vote (classification) or average (regression)
	Stacking	Train a meta-model (blender) on the base learners' outputs to learn the optimal combination
How do we generate base learners that complement each other?	**Bagging**	Train models in parallel on different bootstrap samples of the data
	Boosting	Train models sequentially, each focusing on the errors of the previous ones
	Cascading	Arrange models in a sequence where later models handle harder cases that earlier ones couldn't resolve

Auto Machine Learning

Automated Machine Learning (Auto-ML)

- **Machine learning (ML) crucially relies on human experts to perform the following tasks:**
  - Preprocess and clean the data.
  - Select and construct appropriate features.
  - Select an appropriate model family.
  - Optimize model hyperparameters.
  - Design the topology of neural networks (if deep learning is used).
  - Postprocess machine learning models.
  - Critically analyze the results obtained.
- **What's the goal of Auto-ML?**


Business User
Problem
Data Science
Automation
Optimization
Actionable Insights
Problem Framing
Data Prep & Annotation
Data Ingestion & Management
Partitioning
EDA & Quality Assessment
Feature Engineering
Model Design
Model Tuning
Evaluation & Selection
Risk & Compliance
Software Construction
Deployment
Consumption
Model Maintenance

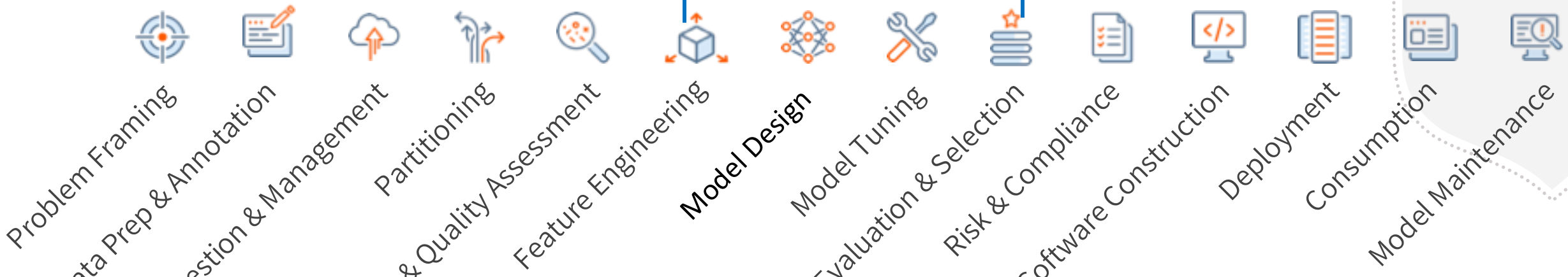
Machine Learning =
Data + ML Expertise + Computation

AutoML = ?

Data + 100X Computation

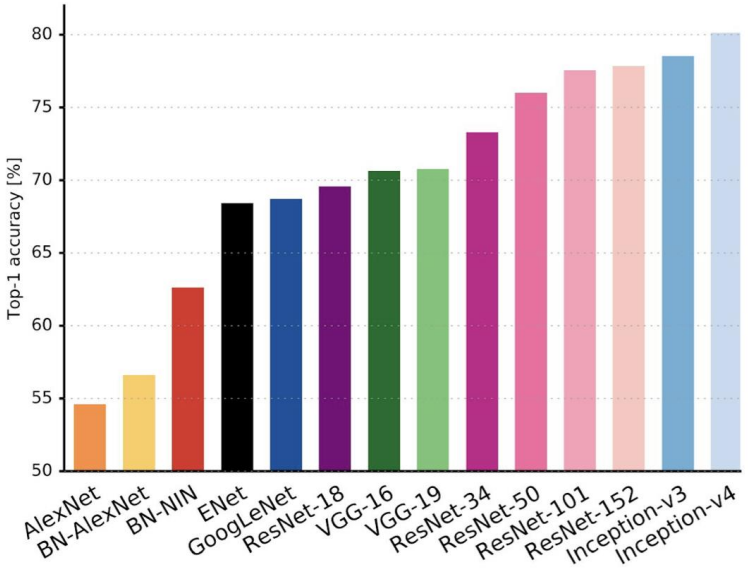
Vast majority of AutoML research and emerging products focused here

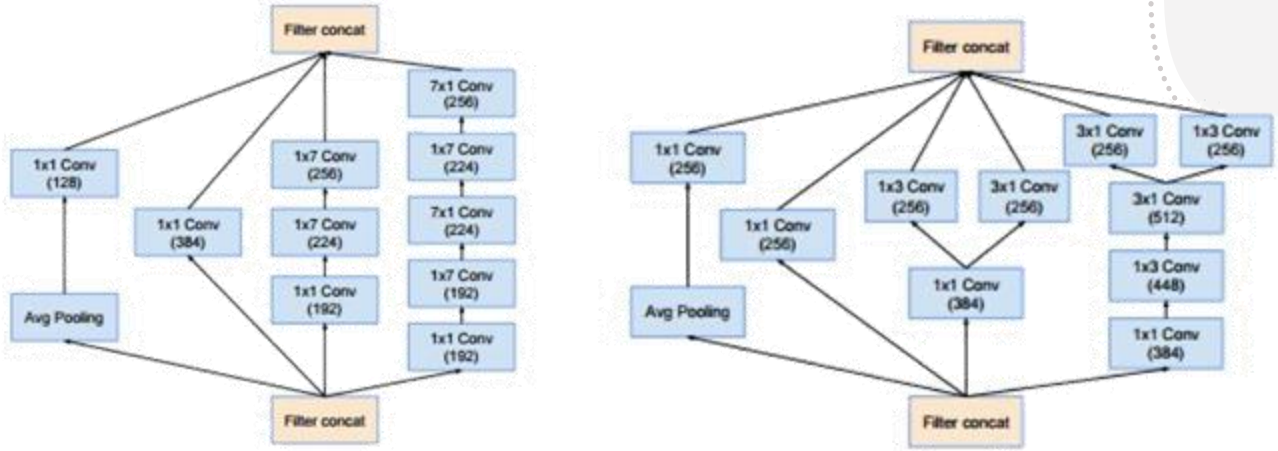
Vast majority of ML research focused here



Automated Model Design

- **Designing neural network architectures is hard**
- **Lots of human efforts go into tuning them**
- **There is not a lot of intuition into how to design them well**
- **Can we try and learn good architectures automatically?**

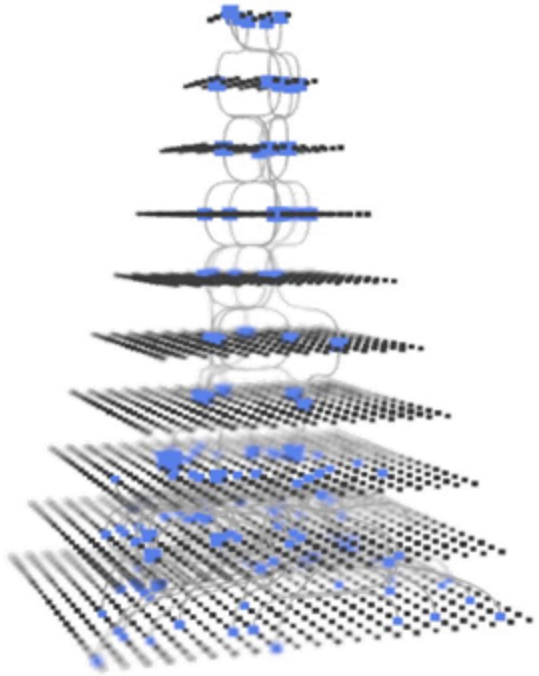




Two layers from the famous Inception V4 computer vision model.

Neural Architecture Search

Controller: proposes ML models



20K times

Iterate to find the most accurate model

Train & evaluate models



References

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